

Heinrich Weber
Lehrbuch der Algebra, Volume II
Current English Translation Draft

Reader note

This current working draft combines the available English translation pieces for Heinrich Weber's *Lehrbuch der Algebra*, Volume II: the preface, §§1–8 of the general group theory opening, and the current continuations on abelian groups, bases, invariants, group characters, reciprocal groups, Lagrange resolvents, linear substitutions, transposed substitutions, projective actions, binary substitutions, absolute invariants, form problems, Galois groups, metacyclic groups, abstract groups, alternating groups, orthogonal groups, fractional linear groups, the icosahedral group, and finite linear fractional groups through §69. The translation is intended as a continuation base for checking and refinement, not as a proofed edition.

Preface to the Second Volume

In accordance with the intention announced in the preface to the first volume, I am now able to place the second volume of my *Textbook of Algebra* before the public. The plan laid down there has been carried out in its essential points. In the applications I have tried to select problems that had already acquired an independent interest in other fields, such as geometry or the theory of functions, and that at the same time bring the principal points of algebraic theory before the reader in as many different ways as possible.

The application of the theory of algebraic numbers has been carried through as far as the theory of cyclotomic numbers. If life and strength endure, I hope in a continuation of my work to present the further applications to the domain of elliptic functions, which are contained only in part in my book *Elliptic Functions and Algebraic Numbers*.

During the preparation and printing of the second volume as well, I have had beside me the help and counsel of the friends whom I already named in the preface to the first edition. But the first volume has by now also won for itself many new friends who have advanced my work by suggestions and advice. To them all I express my thanks here, and add the request that they may continue to preserve their interest in the work.

Strasbourg, July 1896.
The Author.

Part I. General Group Theory

§1. Definition of Groups

In the first volume, in connection with permutations, we became acquainted with the concept of a group and made important algebraic applications of it. Our next task must therefore be to formulate more generally this concept, so extraordinarily important in all of modern mathematics,

and to become familiar with the laws that govern it. We place the following definition at the head of the discussion.

A system P of objects (elements) of any kind whatsoever becomes a *group* when the following assumptions are satisfied.

1. A rule is given according to which, from a first and a second element of the system, a completely determined third element of the same system is derived.

Symbolically, if a is the first, b the second, and c the third element, one writes

$$ab = c, \quad c = ab,$$

and says that c is *composed* from a and b , while a and b are the *components* of c .

For this composition one does not in general assume the commutative law; that is, ab may differ from ba . On the other hand, one does assume

2. the *associative law*,

that is, whenever a, b, c are any three elements of P , one has

$$(ab)c = a(bc).$$

From this, by complete induction, it follows that one always arrives at the same result when, in any finite sequence of elements of P , $a, b, c, d, \dots$, one first composes two neighboring elements, then again two neighboring elements, and so on until the whole sequence has been reduced to a single element, denoted by $abcd \dots$. Thus, for example,

$$\begin{aligned} abcd &= (ab)cd = [(ab)c]d = (ab)(cd) \\ &= a(bc)d = [a(bc)]d = a[(bc)d] \\ &= ab(cd) = (ab)(cd) = a[b(cd)]. \end{aligned}$$

§2. Intersection of two groups

Cosets do not themselves have the defining properties of a group. For, in order that the composition of two elements of Q_1 should again yield an element of Q_1 , one would need, for example,

$$a_1 b a_2 b = a_3 b,$$

and hence

$$a_1 b a_2 = a_3, \quad b = a_1^{-1} a_3 a_2^{-1}.$$

Thus b would, contrary to the assumption, have to belong to Q . The designation *coset* is therefore to be understood only in an inexact or analogical sense.

Two divisors Q, Q' of a group P always have the element 1 in common. They may, however, have further elements in common, and these common elements form a group, which we denote by R . For if a and b belong both to Q and to Q' , then by the group property of Q and Q' the same is true of the product ab ; therefore R is a group. This common group is called the greatest common divisor of Q and Q' , or, with a geometric turn of phrase, the *intersection* of Q and Q' .

§3. Normal subgroups

If Q is a divisor of P that is identical with all of its conjugate divisors, then Q is called a *normal subgroup* of P .

The group consisting only of the single element 1 is a normal subgroup of every group. We obtain another normal subgroup by taking the greatest common divisor R of all divisors conjugate to some divisor Q of P .

Indeed, it has already been proved that R , as the intersection of several divisors of P , is itself a group. Now let

$$Q, Q', Q'', \dots$$

be the divisors conjugate to Q , and let b be any element of P . Then the system of groups

$$b^{-1}Qb, b^{-1}Q'b, b^{-1}Q''b, \dots$$

is not different from the original system. If R is the intersection of the first system, then $b^{-1}Rb$ is the intersection of the second system, and consequently $R = b^{-1}Rb$; that is, R is a normal subgroup of P .

If N is a normal subgroup of a group P and b is any element of P , then

$$b^{-1}Nb = N, \quad \text{or equivalently} \quad Nb = bN.$$

If P has order n , N has order ν , and index μ , so that $n = \mu\nu$, then we may choose the μ elements $1, b_1, \dots, b_{\mu-1}$ in such a way that the decomposition of P into cosets is

$$P = N + Nb_1 + Nb_2 + \dots + Nb_{\mu-1} = N + b_1N + b_2N + \dots + b_{\mu-1}N.$$

§4. Composition of subsets

Let A and B be any two collections of elements of a group P (whether groups or not). By the symbolic product AB we understand the set of all elements obtained by composing each element a of A with each element b of B .

The two equations

$$AA = A, \quad AB = BA,$$

where A is fixed and B is an arbitrary subset of P , express that A is a normal subgroup of P . For from the first relation A is itself a divisor of P , and from the second it follows for every element $b \in P$ that

$$b^{-1}Ab = A,$$

that is, A is normal in P .

If A and B are two divisors of P , then one of the two equalities

$$AB = BA, \quad \text{or equivalently} \quad BA = AB,$$

is the necessary and sufficient condition for the product set AB to be a group.

Because of the defining property of normal subgroups, N commutes with every b , and therefore

$$b_\nu N = Nb_\nu,$$

so that

$$Nb_\nu \cdot Nb_\mu = NN \cdot b_\nu b_\mu.$$

Since $b_\nu b_\mu$ lies in P , the set $Nb_\nu b_\mu$ coincides with one of the cosets occurring in the decomposition above; and because $NN = N$, it follows that if we write $Nb_\nu b_\mu = M$, then

$$NM = M.$$

Thus property (1) in the definition of a group has been verified for the composition of cosets. That the associative law also holds was already noted. It remains only to verify the analogue of cancellation.

§5. Multiple isomorphism

These quotient groups are related to one another in the same way as isomorphic groups in the ordinary sense, except that to each element A there may correspond not just one element a , but several. This gives occasion to extend the notion of isomorphism by the following definition.

A group P with elements $a, b, c, \dots$ is said to be *multiply isomorphic* to a group Q with elements $A, B, C, \dots$ if the two groups can be put into correspondence in such a way that to each of the elements $A, B, C, \dots$ there correspond one or more of the elements $a, b, c, \dots$, while every element of Q corresponds to one and only one element of P , and such that whenever a corresponds to A and b corresponds to B , then ab corresponds to AB .

In this way one proves first that to each of the elements of the quotient group there corresponds an entire coset of the original group. Following H"older, one writes such a quotient group in the form P/N . The order of the complementary group is equal to the index of the group N .

We mention in conclusion a theorem about normal subgroups. If N is a divisor of P , and N' is a divisor of N which is at the same time a normal subgroup of P , then N' is also a normal subgroup of N . This is immediate, because for every element $b \in P$ one has

$$b^{-1}N'b = N',$$

and every element of N is also an element of P .

The converse, however, is not valid in general: if N is a normal subgroup of P and N' is a normal subgroup of N , it need not follow that N' is a normal subgroup of P .

§6. Composition series and the theorem of C. Jordan

A group P is called *simple* if it has no normal subgroup other than itself and the identity; it is called *composite* if other normal subgroups are present.

A normal subgroup which is not contained properly in any larger proper normal subgroup of P is called a *maximal normal subgroup* of P . If P_1 is a maximal normal subgroup of P , P_2 is a maximal normal subgroup of P_1 , and so on, then one obtains a chain

$$P, P_1, P_2, \dots, P_\nu = 1,$$

in which each subgroup is a maximal normal subgroup of the preceding one. Such a chain is called a *composition series*; the corresponding indices are its indices of composition.

The important theorem expressing the uniqueness of the composition factors was established by C. Jordan.

Theorem (Jordan). *However a composition series of a group P may be chosen, the sequence of its indices is always the same up to order.*

The proof rests on the quotient groups introduced above. Suppose Q and Q_1 are normal subgroups of P , and that in addition Q_1 is a divisor (hence, by the previous section, a normal subgroup) of Q . Then the quotient group Q/Q_1 is a normal subgroup of P/Q_1 , and the index of Q/Q_1 in P/Q_1 is equal to the index of Q in P .

This reduction is the crucial device in the proof that the composition factors are independent of the chosen composition series, apart from their order.

§7. Further results on composition series

We derive first several important consequences of Jordan's theorem.

Let P be a group and Q a normal subgroup of P , and suppose that P and Q have composition factors

$$A_1, A_2, \dots, A_\lambda \quad \text{and} \quad C_1, C_2, \dots, C_\mu.$$

Then the quotient group P/Q has composition factors

$$A_1, A_2, \dots, A_\lambda, C_\mu^{-1}, \dots, C_2^{-1}, C_1^{-1}.$$

For every composition series of P/Q can be completed to one of P .

Corollary 1.1. If P, Q, R are three groups, each of which is a normal subgroup of the preceding one, and if the index of R in P is a power of a prime number, then the index of R in Q and the index of Q in P are powers of the same prime.

§8. Metacyclic groups

A group P is called *metacyclic* if it possesses a composition series all of whose factors are cyclic.

This definition includes all cyclic groups, and also all groups whose order is a power of a prime number. For every group of prime-power order is, by Sylow's theorem, solvable, hence metacyclic.

Metacyclic groups play an important role in the theory of algebraic equations, since the equations whose groups are metacyclic are solvable by radicals.

Appendix: foundations of group theory

The preceding paragraphs contain the foundations of general group theory as they are used in modern algebra. We have treated the definition of a group, the theory of divisors and cosets, normal subgroups and their properties, many-valued isomorphism, and composition series together with Jordan's theorem.

These concepts are of fundamental importance for the further development of algebra. Upon them is based the theory of abelian groups, which is presented in the following paragraphs of the second section.

Second Section. Abelian groups

§9. Representation of Abelian groups by a basis

We have already called those groups in which the commutative law holds for composition *commutative* or *Abelian* groups. For these groups, which are of particular importance in applications, the laws are much simpler than in general groups, as we now see.

Let S be an Abelian group of order n . We shall show that the elements of S can be represented in the form

$$\theta = A_1^{\alpha_1} A_2^{\alpha_2} \cdots A_\nu^{\alpha_\nu}, \quad (1)$$

where $A_1, \dots, A_\nu$ are suitable elements of the group and the exponents $\alpha_1, \dots, \alpha_\nu$ run through complete residue systems modulo positive integers $a_1, \dots, a_\nu$, the orders of $A_1, \dots, A_\nu$. Such a system $A_1, \dots, A_\nu$ is called a *basis* of the Abelian group.

One sees immediately that every element of the form (1) actually occurs: for example, one obtains A_i by taking $\alpha_i = 1$ and $\alpha_k = 0$ for $k \neq i$.

It remains to prove that every element occurs equally often. Suppose

$$1 = A_1^{h_1} A_2^{h_2} \cdots A_n^{h_n} \quad (2)$$

is a representation of the identity. Then the value of θ in (1) is unchanged if one replaces $\alpha_1, \dots, \alpha_n$ by $\alpha_1 + h_1, \dots, \alpha_n + h_n$. Hence each element θ is represented at least as many times by (1) as the identity is represented by (2).

Conversely, if two exponent systems $\alpha_1, \dots, \alpha_n$ and $\alpha'_1, \dots, \alpha'_n$ yield the same element θ , then

$$1 = A_1^{\alpha_1 - \alpha'_1} A_2^{\alpha_2 - \alpha'_2} \cdots A_n^{\alpha_n - \alpha'_n}. \quad (3)$$

Thus the difference of the two exponent systems gives a representation of the identity. Therefore the number of representations of θ can be no larger than the number of representations of 1. If we denote that number by h , then the total number of possible exponent choices in (1) is $a_1 a_2 \cdots a_n$, while n is the number of distinct group elements. Hence

$$nh = a_1 a_2 \cdots a_n. \quad (4)$$

From formula (4) one obtains:

Proposition 2. If r is a prime dividing the order n of the group S , then S contains an element of order r .

Indeed, one of the orders $a_1, \dots, a_n$ must be divisible by r . If, say, $a_1 = rk$, then A_1^k has order r .

Proposition 3. If $A, B, \dots$ are arbitrary elements of S with orders $a, b, \dots$, then the product $AB \cdots$ again lies in S , and its order divides the least common multiple m of $a, b, \dots$

For

$$(AB \cdots)^m = A^m B^m \cdots = 1,$$

so m is a multiple of the order of $AB \cdots$.

Proposition 4. Suppose the order n of the group is factored as

$$n = ab$$

with a and b relatively prime. Then there exist exactly a elements A whose order divides a , and exactly b elements B whose order divides b , and every element of the group is represented uniquely in the form

$$\theta = AB. \quad (5)$$

To see this, let $\mathcal{A}$ be the set of elements whose order divides a . By Proposition 3 this is a subgroup of S . Similarly the set $\mathcal{B}$ of elements whose order divides b is a subgroup.

Let a' be the order of $\mathcal{A}$, and b' the order of $\mathcal{B}$. Then a' is relatively prime to b : if a prime r divided a' , then by Proposition 2 the subgroup $\mathcal{A}$ would contain an element of order r , and since every element of $\mathcal{A}$ has order dividing a , that prime r would divide a , not b . Similarly b' is relatively prime to a .

Now choose integers x, y such that

$$ax + by = 1. \quad (6)$$

For any element $\theta \in S$ we have

$$\theta = \theta^{ax} \theta^{by}. \quad (7)$$

Since $(\theta^{ax})^b = 1$, the element θ^{ax} lies in $\mathcal{A}$, and similarly $\theta^{by} \in \mathcal{B}$. Therefore every element has a representation

$$\theta = AB. \quad (8)$$

This representation is unique. If $AB = A'B'$, then by raising to the suitable powers $by = 1 - ax$ and $ax = 1 - by$ one obtains $A = A'$ and hence $B = B'$. The number of products AB is therefore $a'b'$, and since every element of S occurs once, we have

$$n = ab = a'b'.$$

Because a is relatively prime to b' and b is relatively prime to a' , it follows that

$$a = a', \quad b = b'.$$

This proves Proposition 4 in full.

If the two subgroups $\mathcal{A}, \mathcal{B}$ are each represented by bases, then formula (8) shows that S itself is represented by a basis, namely by adjoining the basis elements of $\mathcal{A}$ and $\mathcal{B}$.

Repeating this decomposition whenever a and b can themselves be split into relatively prime factors, we reduce matters to the case where the order of the group is a power of a single prime:

$$n = p^\mu. \quad (9)$$

Thus it remains to prove that every Abelian group of prime-power order can be represented by a basis. Let A be an element of maximal order a . Then a is a power of p , and the order of every element of S divides a . Hence for every $\theta \in S$,

$$\theta^a = 1. \quad (10)$$

The elements

$$1, A, A^2, \dots, A^{a-1} \quad (11)$$

are all distinct and form a subgroup of S . If they already exhaust the group, then S is represented by the one-element basis A .

Otherwise, for every $\theta \in S$ there exists a least positive integer b such that

$$\theta^b = A^t \quad (12)$$

for some integer t . This b divides a , hence is a power of p , and it also divides the order of θ . From (12) one finds that t must be divisible by b : writing $a = qb + b'$ with $0 \leq b' < b$, one checks that if $b' \neq 0$ then $\theta^{b'}$ would already be a power of A , contradicting the minimality of b .

Writing $a/b = k$ and setting

$$B = \theta A^{-tk}, \quad (13)$$

we obtain

$$B^b = 1, \quad (14)$$

and b is the least exponent with this property. Thus the elements

$$A^{\alpha_1} B^{\alpha_2}, \quad \alpha_1 = 0, 1, \dots, a-1, \quad \alpha_2 = 0, 1, \dots, b-1, \quad (15)$$

are all distinct and form a subgroup S' represented by the basis A, B . If $S' = S$, we are done; otherwise the construction continues inductively.

Proceeding by induction, one obtains a subgroup $S_{\nu-1}$ with basis

$$S_{\nu-1} = A_1^{\alpha_1} A_2^{\alpha_2} \cdots A_{\nu-1}^{\alpha_{\nu-1}}, \quad (16)$$

such that:

1. the orders satisfy

$$a_1 \geq a_2 \geq \cdots \geq a_{\nu-1},$$

2. for every element $\theta \in S$, the power $\theta^{a_{\nu-1}}$ lies in $S_{\nu-1}$, and the corresponding exponents are all divisible by $a_{\nu-1}$.

If $S_{\nu-1} \neq S$, choose $\theta \in S$ so that the least positive exponent a_ν with $\theta^{a_\nu} \in S_{\nu-1}$ is as large as possible. Then $a_\nu \leq a_{\nu-1}$, and a_ν is again a power of p . Writing

$$\theta^{a_\nu} = A_1^{h_1 a_\nu} A_2^{h_2 a_\nu} \cdots A_{\nu-1}^{h_{\nu-1} a_\nu}, \quad (17)$$

define

$$A_\nu = \theta A_1^{-h_1} A_2^{-h_2} \cdots A_{\nu-1}^{-h_{\nu-1}}. \quad (18)$$

Then

$$A_\nu^{a_\nu} = 1, \quad (19)$$

and a_ν is the smallest such exponent. Consequently, the elements

$$A_1^{\alpha_1} A_2^{\alpha_2} \cdots A_\nu^{\alpha_\nu}, \quad 0 \leq \alpha_i < a_i, \quad (20)$$

are all distinct and form a larger subgroup S_ν , again satisfying the inductive conditions.

Continuing this process, one eventually obtains the whole group S represented by a basis. This proves the theorem that every finite Abelian group can be represented by a basis.

It follows at once that every Abelian group is metacyclic. Indeed, if

$$\theta = A_1^{\alpha_1} A_2^{\alpha_2} \cdots A_\nu^{\alpha_\nu},$$

and if p is a prime dividing the order a_1 of A_1 , then the elements

$$\theta' = A_1^{p\alpha_1} A_2^{\alpha_2} \cdots A_\nu^{\alpha_\nu}$$

form a subgroup of index p , represented by the basis $A_1^p, A_2, \dots, A_\nu$. Repeating this procedure shows that S admits a chain of subgroups of prime index, hence is metacyclic.

§10. The invariants of Abelian groups

The proof just given for the existence of a basis also provides a method for finding one in which the orders of the basis elements are prime powers. Nevertheless, the same Abelian group can admit different bases.

For example, let A and B have relatively prime orders a and b . Then they determine a two-element basis with group

$$A^\alpha B^\beta, \quad \alpha = 0, 1, \dots, a-1, \quad \beta = 0, 1, \dots, b-1. \quad (1)$$

But the same group can also be represented by the one-element basis AB , namely in the form

$$(AB)^s, \quad s = 0, 1, \dots, ab-1. \quad (2)$$

Indeed, given arbitrary α, β , one can determine s modulo ab so that

$$s \equiv \alpha \pmod{a}, \quad s \equiv \beta \pmod{b},$$

and then (2) reproduces (1).

Even so, there is a precise sense in which the structure of the basis is completely determined by the group itself.

Theorem 0.2. Suppose

$$A_1, A_2, \dots, A_\nu$$

with orders

$$a_1, a_2, \dots, a_\nu$$

and

$$B_1, B_2, \dots, B_\mu$$

with orders

$$b_1, b_2, \dots, b_\mu$$

are two bases of the same Abelian group S of order n . Let p be a prime dividing n , and write

$$a_i = p^{\alpha_i} \bar{a}_i, \quad b_j = p^{\beta_j} \bar{b}_j,$$

where $\bar{a}_i, \bar{b}_j$ are not divisible by p . Then the prime powers p^{α_i} greater than 1 occur among the p^{β_j} , and conversely.

To prove this, order the basis elements so that

$$a_1 \leq a_2 \leq \dots \leq a_\nu, \quad b_1 \leq b_2 \leq \dots \leq b_\mu. \quad (3)$$

Let m be the least common multiple of $a_1, \dots, a_\nu$. Since the $\bar{a}_i$ are not divisible by p , the number m is also not divisible by p .

For an arbitrary element

$$\theta = A_1^{\alpha_1} A_2^{\alpha_2} \dots A_\nu^{\alpha_\nu}, \quad (4)$$

one then has

$$\theta^{p^{\alpha_\nu} m} = 1.$$

Applying this to the basis elements $B_1, \dots, B_\mu$, one sees that every b_j divides $p^{\alpha_\nu} m$. Hence p^{β_j} divides p^{α_ν} , and the least common multiple of the b_j is again m . Reversing the rôles of the two bases, one obtains the opposite divisibility, and therefore

$$p^{\alpha_\nu} = p^{\beta_\mu}. \quad (5)$$

Assume inductively that

$$p^{\alpha_1} = p^{\beta_1}, \quad p^{\alpha_2} = p^{\beta_2}, \quad \dots, \quad p^{\alpha_{s-1}} = p^{\beta_{s-1}} \quad (6)$$

have already been proved. Then for any element θ ,

$$\theta^{p^{\alpha_s} m} = A_1^{\alpha_1 p^{\alpha_s} m} A_2^{\alpha_2 p^{\alpha_s} m} \dots A_\nu^{\alpha_\nu p^{\alpha_s} m}. \quad (7)$$

Counting the number of distinct elements of this form yields

$$q = \frac{p^{\alpha_1}}{p^{\alpha_s}} \cdot \frac{p^{\alpha_2}}{p^{\alpha_s}} \dots \frac{p^{\alpha_{s-1}}}{p^{\alpha_s}}. \quad (9)$$

If one now expresses the same elements in the B -basis and assumes

$$p^{\alpha_s} > p^{\beta_s}, \quad (10)$$

then the counting argument forces p^{β_s} to divide p^{α_s} , which contradicts (10) unless

$$p^{\alpha_s} = p^{\beta_s}. \quad (14)$$

This proves Theorem 0.2.

The prime powers occurring in the orders of a basis are therefore independent of the choice of basis. We call them the *invariants* of the Abelian group. Their product is equal to the order n of the group.

By §9 we can choose a basis whose element orders are prime powers. Hence those orders are exactly the invariants and are completely determined by the group.

That the invariants completely determine the group as well follows from the next theorem.

Theorem 0.3. Two Abelian groups with the same invariants are isomorphic, and isomorphic Abelian groups have the same invariants.

Indeed, if

$$\theta = A_1^{\alpha_1} A_2^{\alpha_2} \dots A_\nu^{\alpha_\nu}$$

are the elements of one group S , and

$$\theta' = B_1^{\alpha_1} B_2^{\alpha_2} \dots B_\nu^{\alpha_\nu}$$

those of another group S' , with the same exponent ranges modulo the same basis orders $a_1, \dots, a_\nu$, then we obtain an isomorphism by letting θ correspond to θ' whenever the exponent systems are identical.

Conversely, if two groups are isomorphic, then the images of a basis of one group form a basis of the other, and therefore the invariants must agree in both.

§11. Group characters

A principal problem in the theory of Abelian groups is to determine all divisors of an Abelian group. The solution of this problem is substantially facilitated by introducing the concept of *group characters*, which is also important in many other respects.

Let S be an Abelian group of order n , whose elements we denote by A , and let

$$A_1, A_2, \dots, A_\nu \quad (1)$$

be the elements of a basis of S , of orders

$$a_1, a_2, \dots, a_\nu.$$

Then every element A is represented, and represented uniquely, in the form

$$A = A_1^{\alpha_1} A_2^{\alpha_2} \dots A_\nu^{\alpha_\nu}, \quad (2)$$

where

$$0 \leq \alpha_1 < a_1, \quad 0 \leq \alpha_2 < a_2, \quad \dots, \quad 0 \leq \alpha_\nu < a_\nu. \quad (3)$$

The exponents $\alpha_1, \alpha_2, \dots, \alpha_\nu$, or any integers congruent to them modulo $a_1, a_2, \dots, a_\nu$, are called the *indices* of the element A . Then one has:

1. The indices of a product AA' are obtained by adding the corresponding indices of the two factors.

If we assign to each element A some numerical value, we may call this assignment a function of A . Such a function $\chi(A)$ will be called a *group character* if for every two elements $A, A' \in S$ it satisfies

$$\chi(AA') = \chi(A)\chi(A'). \quad (4)$$

Hence it also satisfies the more general multiplicative law

$$\chi(A'A''A''' \dots) = \chi(A')\chi(A'')\chi(A''') \dots$$

Two such functions χ and χ_1 are regarded as distinct if there exists at least one element A for which $\chi(A) \neq \chi_1(A)$. We now determine these characters more closely.

Setting $A' = 1$ in (4), one obtains

$$\chi(A) = \chi(A)\chi(1). \quad (5)$$

Excluding once and for all the degenerate case in which all $\chi(A) = 0$, one gets:

$$\chi(1) = 1. \quad (6)$$

2. Every character takes the value 1 on the identity element.

Setting $A' = A$ in (4), we obtain

$$\chi(A^2) = [\chi(A)]^2,$$

and by complete induction, for every exponent h ,

$$\chi(A^h) = [\chi(A)]^h. \quad (7)$$

If a denotes the order of the element A , then with $h = a$ and using (6) one gets

$$[\chi(A)]^a = 1. \quad (8)$$

3. The values of a character $\chi(A)$ are roots of unity whose orders divide the order of A .

We can now determine all characters easily. Set

$$\chi(A_1) = \omega_1, \quad \chi(A_2) = \omega_2, \quad \dots, \quad \chi(A_\nu) = \omega_\nu. \quad (9)$$

Then

$$\omega_1^{a_1} = 1, \quad \omega_2^{a_2} = 1, \quad \dots, \quad \omega_\nu^{a_\nu} = 1, \quad (10)$$

and from (2) and (4),

$$\chi(A) = \omega_1^{\alpha_1} \omega_2^{\alpha_2} \dots \omega_\nu^{\alpha_\nu}. \quad (11)$$

Conversely, whenever $\omega_1, \omega_2, \dots, \omega_\nu$ is any solution of (10), formula (11) defines a function of A that satisfies (4), and hence is a character of the group.

Each of the equations (10) has $a_1, a_2, \dots, a_\nu$ roots, and if all root choices are combined with one another, one obtains

$$a_1 a_2 \dots a_\nu = n$$

possible systems. These lead to n distinct characters $\chi(A)$. Indeed, if another choice

$$\omega'_1, \omega'_2, \dots, \omega'_\nu$$

yields the same function for all A , then taking $\alpha_1 = 1, \alpha_2 = \dots = \alpha_\nu = 0$ shows $\omega_1 = \omega'_1$, and similarly for the others.

Thus:

4. There exist exactly n distinct characters of an Abelian group of order n , all given by formula (11).

Let $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_\nu$ be primitive roots of the equations (10). Then we may write

$$\omega_1 = \varepsilon_1^{\beta_1}, \quad \omega_2 = \varepsilon_2^{\beta_2}, \quad \dots, \quad \omega_\nu = \varepsilon_\nu^{\beta_\nu}, \quad (12)$$

where $\beta_1, \dots, \beta_\nu$ are taken from complete residue systems modulo $a_1, \dots, a_\nu$. Then all n characters are given by

$$\chi(A) = \varepsilon_1^{\alpha_1 \beta_1} \varepsilon_2^{\alpha_2 \beta_2} \dots \varepsilon_\nu^{\alpha_\nu \beta_\nu}. \quad (13)$$

Among these occurs the character obtained by setting all $\beta_i = 0$; it has value 1 on every element A , and we call it the *unit character*.

The n characters of S can themselves be combined into an Abelian group, indeed into one isomorphic to S .

For by (13), each character is determined by a system of integers

$$\beta_1, \beta_2, \dots, \beta_\nu$$

taken modulo $a_1, \dots, a_\nu$, and this same system is the index system of a definite element $B \in S$, namely

$$B = A_1^{\beta_1} A_2^{\beta_2} \dots A_\nu^{\beta_\nu}. \quad (14)$$

Thus every element $B \in S$ corresponds to a definite character, which we denote by $\chi_B(A)$, or simply χ_B when the variable A is omitted.

If B' is a second element of S , with indices $\beta'_1, \beta'_2, \dots, \beta'_\nu$, then its corresponding character $\chi_{B'}$ is defined likewise. If now $\chi_B \chi_{B'}$ denotes the character given on each A by

$$\chi_B \chi_{B'}(A) = \varepsilon_1^{\alpha_1(\beta_1 + \beta'_1)} \varepsilon_2^{\alpha_2(\beta_2 + \beta'_2)} \dots \varepsilon_\nu^{\alpha_\nu(\beta_\nu + \beta'_\nu)}, \quad (15)$$

then the characters are thereby combined into a group isomorphic to S . The identity element of this character group is the unit character.

Accordingly, the character group can itself be represented by a basis, obtained by setting

$$\chi_1(A) = \varepsilon_1^{\alpha_1}, \quad \chi_2(A) = \varepsilon_2^{\alpha_2}, \quad \dots, \quad \chi_\nu(A) = \varepsilon_\nu^{\alpha_\nu}. \quad (16)$$

Then every character has the form

$$\chi_B = \chi_1^{\beta_1} \chi_2^{\beta_2} \cdots \chi_\nu^{\beta_\nu}, \quad (17)$$

and if $B, B' \in S$, then

$$\chi_B \chi_{B'} = \chi_{BB'}. \quad (18)$$

If χ_1, χ_2 are any two characters and $\chi_1 \chi_2$ their composite, then for every A ,

$$\chi_1(A) \chi_2(A) = (\chi_1 \chi_2)(A). \quad (19)$$

We may summarize this as:

5. The characters of a group S can be combined into a group isomorphic to S .

One has also, immediately from definition (13),

$$\chi_B(A) = \chi_A(B). \quad (20)$$

Finally there is the following theorem.

6. If A runs through all elements of the group S , then for a fixed character χ ,

$$\sum_A \chi(A) = \begin{cases} n, & \text{if } \chi \text{ is the unit character,} \\ 0, & \text{otherwise.} \end{cases} \quad (21)$$

Likewise, if the element A is held fixed and χ runs through all characters, then

$$\sum_\chi \chi(A) = \begin{cases} n, & \text{if } A = 1, \\ 0, & \text{otherwise.} \end{cases} \quad (22)$$

When χ is the unit character, or A is the identity, these formulas are evident, since each of the n summands is then equal to 1. If χ is not the unit character, there exists $B \in S$ such that $\chi(B) \neq 1$. Writing

$$\sum_A \chi(A) = \sigma,$$

and multiplying by $\chi(B)$, one obtains

$$\sum_A \chi(AB) = \sigma \chi(B).$$

But as A runs through the whole group, so does AB , hence the left side is also σ . Therefore

$$\sigma[1 - \chi(B)] = 0,$$

and since $\chi(B) \neq 1$, we get $\sigma = 0$, proving (21). Formula (22) follows similarly, and also immediately from (20).

§12. Divisors of an Abelian group. Reciprocal groups

A divisor T of an Abelian group S is, as already remarked above, always a normal subgroup. Hence by §4 there exists a complementary group

$$S/T,$$

whose elements are the cosets of T , namely

$$T, \quad TA', \quad TA'', \quad \dots, \quad (1)$$

where $A', A'', \dots$ are certain elements of S . If t is the order of T , then the number of such cosets, that is, the order of the group S/T , is equal to the index of T in S , namely

$$\frac{n}{t}.$$

This group S/T is itself Abelian, because

$$TA' \cdot TA'' = TA'A'', \quad TA'' \cdot TA' = TA''A',$$

and these are equal.

Let us now determine the characters of the group S/T . Denote its elements by

$$T_0, T_1, T_2, \dots, T_j, \quad (2)$$

and let one of its characters be written $\xi(T_i)$.

From this function $\xi(T)$ we can derive a function $\xi(A)$ on the elements of S by setting

$$\xi(A_i) = \xi(T_i) \quad (3)$$

whenever A_i is any element belonging to the coset T_i . For the elements of the group T itself one then has $\xi(A) = 1$.

But this function $\xi(A)$ is among the characters of S . For if $A_i \in T_i$ and $A_k \in T_k$, then $A_i A_k \in T_i T_k$, and hence

$$\xi(A_i)\xi(A_k) = \xi(T_i)\xi(T_k) = \xi(T_i T_k) = \xi(A_i A_k). \quad (4)$$

This is precisely the defining property of a character of S .

Conversely, suppose one of the characters $\chi = \xi$ of S takes the same value on every element of T . Since the identity belongs to T , this common value can only be 1. If now A_i is any element of the coset T_i , and A runs through the whole group T , then AA_i runs through the coset T_i . Since $\xi(A) = 1$, one has

$$\xi(AA_i) = \xi(A_i). \quad (5)$$

That is, $\xi(A)$ has one and the same value on all elements of a fixed coset T_i , and may therefore be regarded as a function of T_i , denoted $\xi(T_i)$.

Since whenever $A_i \in T_i$ and $A_k \in T_k$, the product $A_i A_k \in T_i T_k$, it follows that

$$\xi(T_i)\xi(T_k) = \xi(T_i T_k). \quad (6)$$

So $\xi(T_i)$ is indeed a character of the quotient group S/T .

Therefore there are exactly j such characters $\xi(A)$, namely those that take the value 1 on every element of T . These j characters form, under composition of characters, a group; for if $\xi_1(A) = 1$ and $\xi_2(A) = 1$ on T , then also $\xi_1 \xi_2(A) = 1$. Since these functions ξ can equally well be viewed as the characters of the quotient group S/T , it follows from §11, theorem 5, that the group of the ξ is isomorphic to S/T .

Hence:

7. If an Abelian group S has a divisor T of order t and index j , then among the characters of S there are exactly j , and no more, that take the value 1 on all elements of T ; and for every element of S not contained in T , at least one of these characters takes a value different from 1. These characters form a group isomorphic to S/T .

Since every character χ_B corresponds to a definite element $B \in S$, if χ_B runs through the group of characters belonging to T , then B runs through a group of order j , isomorphic to the group of the χ_B , and therefore also to S/T . We denote this group by U and call it the *group reciprocal to T* .

Again, one should note that in general the notion of reciprocal group depends on the chosen basis and the chosen roots of unity ε , so it does not belong intrinsically to the group S and the subgroup T alone.

The reciprocal group of T is characterized by the following theorem.

8. Let A run through the elements of a divisor $T \subset S$. Then the elements $B \in S$ satisfying

$$\chi_B(A) = 1 \tag{7}$$

for every $A \in T$ are precisely the elements of the group U reciprocal to T , whose order is equal to the index of T .

By §11, (20), the group T is in turn the reciprocal group to U ; the relation between the two groups is therefore mutual.

One also has the following theorem.

9. If T' is a divisor of T , then conversely the reciprocal group U of T is a divisor of the group U' reciprocal to T' .

For every element $A' \in T'$ also lies in T . Hence, if B runs through U , then $\chi_B(A') = 1$, and therefore B must occur in the group U' .

If one selects from the totality of characters χ an arbitrary finite number

$$\xi_1, \xi_2, \dots, \xi_k,$$

whether or not they form a group, then all elements A satisfying the k conditions

$$\xi_1(A) = 1, \quad \xi_2(A) = 1, \quad \dots, \quad \xi_k(A) = 1 \tag{8}$$

form a group, because from $\xi_i(A) = 1$ and $\xi_i(A') = 1$ it follows that $\xi_i(AA') = 1$.

If the characters $\xi_1, \dots, \xi_k$ do not already form a group, then there follow from equations (8) additional equations

$$\xi_{k+1}(A) = 1, \quad \dots$$

so that $\xi_1, \dots, \xi_k$ may be completed to a group.

10. It follows from theorem 7 that in this way one can obtain all possible divisors of S .

§4. Cyclic equations

163. Reduction of Abelian equations to cyclic equations

We have already seen above that in a transitive Abelian group, besides the identity, no permutation occurs that leaves any symbol fixed. From this one obtains another theorem.

Assume that a permutation π of such a group is decomposed into its cycles, and let the cycle with the fewest elements be an r -cycle. Then π^r leaves the elements of this cycle fixed and therefore must be the identity permutation. But from $\pi^r = 1$ it follows that, if other cycles are present, they too must have r elements. Thus:

Theorem. A permutation of a transitive Abelian group contains only cycles of the same length.

If m is the degree of the group, then the number r of elements in one cycle must divide m , and if $m = rs$, then s is the number of r -cycles of which π consists.

Now let P be the group of an Abelian equation $F(x) = 0$, and choose any non-identity permutation π in P . Decompose it into cycles

$$\pi = \gamma_1 \gamma_2 \cdots \gamma_s, \quad (1)$$

where each cycle γ refers to r roots of the equation $F(x) = 0$. Arrange these roots so that

$$\begin{aligned} \gamma_1 &= (\alpha, \alpha_1, \alpha_2, \dots, \alpha_{r-1}), \\ \gamma_2 &= (\beta, \beta_1, \beta_2, \dots, \beta_{r-1}), \\ &\dots, \\ \gamma_s &= (\delta, \delta_1, \delta_2, \dots, \delta_{r-1}). \end{aligned} \quad (2)$$

If π_i is any permutation of P , then because of commutativity one has

$$\pi^{-1} \pi_i \pi = \pi_i, \quad \text{that is,} \quad \pi_i \pi = \pi \pi_i. \quad (3)$$

By the theorem on the formation of the permutations $\pi^{-1} \pi_i \pi$ (§154, no. 6), it follows that γ_1 cannot be changed when the permutations π are carried out. But since γ_1 is completely determined inside the cycles of π , apart from its ordering and its initial element, it follows that under any permutation π_i in P the elements belonging to one cycle γ cannot be separated from one another; they can only be cyclically permuted among themselves, and the cycles can moreover be permuted among one another.

Hence, if $s > 1$, the group is imprimitive.

A rational function of the arguments $\alpha, \alpha_1, \dots, \alpha_{r-1}$ that does not change its value when the α are subjected to the cyclic permutation γ_1 and its repetitions is called a *cyclic function* of the α . If we denote by $\omega, \omega_1, \dots, \omega_{s-1}$ the conjugate values of ω , setting for instance

$$\begin{aligned} \omega &= \psi(\alpha, \alpha_1, \dots, \alpha_{r-1}), \\ \omega_1 &= \psi(\beta, \beta_1, \dots, \beta_{r-1}), \quad \dots, \end{aligned} \quad (4)$$

then these quantities are, by §158, the roots of an irreducible equation of degree s ,

$$\Phi(\omega) = 0, \quad (5)$$

whose group is obtained by determining the permutations induced on the quantities (4) by the permutations coming from P .

But the permutation of the quantities (4) induced by $\pi_i \pi_j$ is obtained by composing the permutations induced separately by π_i and π_j ; hence the group of permutations of (4) is itself commutative.

Therefore $\Phi(\omega) = 0$ is an Abelian equation of degree s . Its first factor $F(x, \omega)$ has the roots $\alpha, \alpha_1, \dots, \alpha_{r-1}$, and the group of the equation

$$F(x, \omega) = 0$$

consists only of the permutations belonging to the period of the cyclic permutation γ_1 .

We shall call an equation whose group consists of a single cycle and its repetitions a *cyclic equation*. Thus cyclic equations are the simplest special case of Abelian equations.

We have therefore proved that the solution of every Abelian equation is reduced to the solution of an Abelian equation of lower degree together with the solution of a collection of cyclic equations.

One can apply the same theorem again to the auxiliary equation of degree s , and continue in this way until that auxiliary equation is reduced to degree one.

Thus we obtain the result:

Theorem. The solution of an Abelian equation can always be reduced to the solution of a sequence of cyclic equations whose degrees divide the degree of the given equation.

It is not necessary to include irreducibility in the definition of cyclic equations. We may therefore formulate the definition in general as follows.

An equation of degree m , $F(x) = 0$, with m distinct roots is called a *cyclic equation* over the field K if its roots $\alpha, \alpha_1, \alpha_2, \dots, \alpha_{m-1}$ are not rational, but can be arranged in such a way that the cyclic functions of the roots are rational over K .

If therefore

$$\pi = (\alpha, \alpha_1, \alpha_2, \dots, \alpha_{m-1}), \quad (6)$$

then every function belonging to K and invariant under the permutations of the cyclic group

$$C = 1, \pi, \pi^2, \pi^3, \dots, \pi^{m-1} \quad (7)$$

must be rational.

The Galois group of a cyclic equation is either the whole period C , in which case the equation is irreducible, or else it is a subgroup of C ; in the latter case, if e and f are two integer divisors of m with $m = ef$, then this group consists of the permutations

$$C^e = 1, \pi^e, \pi^{2e}, \dots, \pi^{(f-1)e}, \quad (8)$$

and the cyclic equation $F(x) = 0$ splits into e factors of degree f .

For let

$$t = \psi(\alpha, \alpha^e, \alpha^{2e}, \dots, \alpha^{(f-1)e})$$

be a function belonging to the group C , and assume it is equal to a quantity in K , which we denote by a .

Then on the rational equation

$$F(x) = F(x, a)F(x, a_1) \cdots F(x, a_{e-1})$$

no permutation not contained in C can act, and consequently the group of the equation can contain no substitutions other than those occurring in C .

If now π itself occurs in the group of the equation, then the group is identical with C , and since C is transitive, the equation is irreducible.

If, however, $\pi^{e'}$ is the smallest power of π occurring in the group of the equation, then $C^{e'}$ is the group. If $e' < m$, then $C^{e'}$ is intransitive, and the equation is reducible.

The same reduction can also be applied to cyclic equations whose degree is not prime. If namely $m = ef$ is any factorization of m into two factors, then the permutation π^e splits into e cycles γ , each with f elements:

$$\gamma = (\alpha, \alpha^e, \alpha^{2e}, \dots, \alpha^{(f-1)e}). \quad (9)$$

Now determine a function ψ belonging to the first of these cycles, and set

$$\begin{aligned}\eta &= \eta_0 = \psi(\alpha, \alpha^e, \alpha^{2e}, \dots, \alpha^{(f-1)e}), \\ \eta_1 &= \psi(\alpha_1, \alpha^{e+1}, \alpha^{2e+1}, \dots, \alpha^{(f-1)e+1}), \\ &\dots, \\ \eta_{e-1} &= \psi(\alpha_{e-1}, \alpha_{2e-1}, \alpha_{3e-1}, \dots, \alpha_{m-1}).\end{aligned}\tag{10}$$

These quantities are all distinct, yet under application of the permutation π they pass cyclically into one another. Hence their cyclic functions, and therefore also their symmetric functions, are rational. So they are the roots of a cyclic equation of degree e , while $F(x)$ splits into e factors of degree f ,

$$F(x) = F(x, \eta)F(x, \eta_1) \cdots F(x, \eta_{e-1}),$$

each of which gives a cyclic equation of degree f for the roots of one of the cycles γ .

Indeed, if for example

$$F_1 = (x - \alpha)(x - \alpha^e) \cdots (x - \alpha^{(f-1)e}),$$

then this function admits the permutations in the period of γ . It also admits the permutations of the group that becomes the Galois group of our equation after adjoining η , a group that can consist only of powers of $\gamma_0, \gamma_1, \dots, \gamma_{e-1}$.

Hence, by Lagrange's theorem (§155, equation (2)), F_1 is rationally expressible in terms of η , so $F_1 = F(x, \eta)$. But the equation

$$F(x, \eta) = 0$$

is again cyclic over $K(\eta)$, since the cyclic functions of its roots belong to that field.

Like all Abelian equations, cyclic equations have the property that every root is rationally expressible in terms of every other. Here these expressions can be arranged in the following cyclic manner. If $\alpha, \alpha_1, \alpha_2, \dots, \alpha_{m-1}$ are the roots of the cyclic equation $F(x) = 0$, then the entire function of degree $m - 1$ in x

$$\Psi_m \left(x + \frac{F'(x)}{\Psi'_m(x)} \right) = \Psi_m$$

is unchanged by the permutation π and therefore belongs to K . If in it we set

$$x = \alpha, \alpha_1, \dots, \alpha_{m-1}$$

and introduce the symbol θ , then it follows that

$$\alpha_1 = \theta(\alpha), \quad \alpha_2 = \theta(\alpha_1), \quad \dots, \quad \alpha_{m-1} = \theta(\alpha_{m-2}), \quad \alpha = \theta(\alpha_{m-1}), \tag{11}$$

and this holds whether $F(x)$ is reducible or irreducible, provided only that $F(x)$ and $F'(x)$ have no common factor.

The solution of cyclic equations of degree m has thus been made dependent on the solution of cyclic equations whose degrees are the prime factors of m . If for example m is a power of 2, then the solution is carried out by a sequence of square roots. In this way Gauss first treated the cyclotomic equations.

Finally we add the remark, important for what follows, that for prime degree the notions of normal equation and cyclic equation coincide. Certainly a cyclic equation of prime degree is irreducible and hence a normal equation.

Conversely, if the degree n of a normal equation, which is also the degree of its group, is prime, and if π is any non-identity permutation of this group, then the degree of π , which must divide n , is equal to n ; hence

$$\pi^n = 1, \quad \pi, \pi^2, \dots, \pi^{n-1}$$

constitute the group of the equation, and the equation is cyclic.

§5. Lagrange resolvents

164. The method of solution for cyclic equations

The method by which we shall now learn to solve cyclic equations applies uniformly to prime degrees and to composite degrees. One uses for this purpose certain expressions known under the name of *Lagrange resolvents*, which are of great use in all investigations concerning the algebraic solution of equations.

Let $F(x) = 0$ be an equation with roots

$$\alpha, \alpha_1, \alpha_2, \dots, \alpha_{m-1}.$$

Let ε be any m th root of unity, and introduce the notation

$$(\varepsilon, \alpha) = \alpha + \varepsilon\alpha_1 + \varepsilon^2\alpha_2 + \dots + \varepsilon^{m-1}\alpha_{m-1}. \quad (1)$$

These sums are what one calls the *Lagrange resolvents*. If these functions are known for all m th roots of unity ε , then the equation itself is solved; for by §133, equation (6),

$$\sum_{\varepsilon} \varepsilon^{-k} = m \quad \text{or} \quad 0, \quad (2)$$

according as k is divisible by m or not, and hence

$$m\alpha_k = \sum_{\varepsilon} (\varepsilon, \alpha) \varepsilon^{-k}, \quad (3)$$

where the sums extend over all m th roots of unity ε .

The other roots can be expressed in the same way:

$$m\alpha_k = \sum_{\varepsilon} \varepsilon^{-k} (\varepsilon, \alpha), \quad (4)$$

so that everything is indeed reduced to knowledge of (ε, α) and the functions formed from it.

The sums (2) and (3) can also be written in another fashion, since all m th roots of unity are powers of a primitive one.

If therefore ε denotes a fixed primitive m th root of unity and h runs through a complete system of residues modulo m , say $0, 1, \dots, m-1$, then equations (2), (3) may be written as

$$m\alpha = \sum_h (\varepsilon^h, \alpha), \quad m\alpha_k = \sum_h \varepsilon^{-hk} (\varepsilon^h, \alpha). \quad (5)$$

We first investigate these resolvents as functions of m independent variables α , and for simpler notation we adopt the convention that $\alpha_h = \alpha_k$ whenever $h \equiv k \pmod{m}$.

Then the entire system of variables α is obtained by letting the index h run through a complete residue system modulo m .

For these resolvents the following statements hold.

1. If one applies to the indices of the α the cyclic permutation

$$\pi = (0, 1, \dots, m-1),$$

then (ε, α) passes into $\varepsilon^{-1}(\varepsilon, \alpha)$, and under the permutation π^k it passes into $\varepsilon^{-k}(\varepsilon, \alpha)$.

This follows immediately from the defining formula (1).

Now let ν be any positive exponent and form $(\varepsilon, \alpha)^\nu$. Expand this power by the polynomial theorem, set $\varepsilon^m = 1$, and order the result by powers of ε . One obtains an expression of the form

$$(\varepsilon, \alpha)^\nu = A_0^{(\nu)} + \varepsilon A_1^{(\nu)} + \varepsilon^2 A_2^{(\nu)} + \dots + \varepsilon^{m-1} A_{m-1}^{(\nu)}, \quad (5')$$

where the $A_h^{(\nu)}$ are forms of degree ν with integer coefficients in the variables α , but are independent of ε . Again we stipulate that $A_h^{(\nu)} = A_k^{(\nu)}$ whenever $h \equiv k \pmod{m}$.

From this we prove the following statement.

2. If one applies the cyclic permutation π to the indices of the α , then the indices of the coefficients in $(\varepsilon, \alpha)^\nu$ undergo the cyclic permutation π^ν ; that is, $A_h^{(\nu)}$ passes into $A_{h+\nu}^{(\nu)}$.

To prove this, note that formula (5') remains valid for every m th root of unity ε , including 1. Using (2) one then obtains

$$mA_k^{(\nu)} = \sum_{\varepsilon} \varepsilon^{-k} (\varepsilon, \alpha)^\nu. \quad (6)$$

If we apply the permutation π on the right-hand side, then by statement 1 the term $(\varepsilon, \alpha)^\nu$ is multiplied by $\varepsilon^{-\nu}$, so the whole sum becomes

$$\sum_{\varepsilon} \varepsilon^{-(k+\nu)} (\varepsilon, \alpha)^\nu = mA_{k+\nu}^{(\nu)},$$

which is exactly the assertion.

3. Thus, when the permutation π is applied to the indices of the α , it induces on the indices of the coefficients of the product ordered by powers of ε the permutation determined by the total exponent-weight.

This general principle will be used in the further computation of the Lagrange resolvents.

§37. Linear substitutions and their composition

One of the most effective means for forming groups, and at the same time one that leads to many applications, is given by linear substitutions and their composition. We have already encountered such linear substitutions several times, for example in the second section of the first volume on the occasion of the multiplication law for determinants, and then again in the eleventh section in connection with equivalent numbers.

By a *linear substitution* in n variables we understand a system of equations by which one system of n variables $y_1, y_2, \dots, y_n$ is expressed linearly in terms of another system $x_1, x_2, \dots, x_n$. According to the number of variables we distinguish unary, binary, ternary, quaternary substitutions, and in general we shall call the number of variables the *dimension* of the substitution. For the time being we restrict attention to homogeneous substitutions, so that, if the coefficients are denoted by $a_i^{(\kappa)}$, the substitution is represented by the system

$$y_\kappa = \sum_{i=1}^n a_i^{(\kappa)} x_i \quad (\kappa = 1, 2, \dots, n). \quad (1)$$

Very often the variables themselves are not important, so that such a substitution is sufficiently characterized by its coefficients $a_i^{(\kappa)}$. One sometimes uses a single letter, say A , to denote a substitution, setting

$$A = \begin{pmatrix} a_1^{(1)} & a_2^{(1)} & \cdots & a_n^{(1)} \\ a_1^{(2)} & a_2^{(2)} & \cdots & a_n^{(2)} \\ \vdots & \vdots & \ddots & \vdots \\ a_1^{(n)} & a_2^{(n)} & \cdots & a_n^{(n)} \end{pmatrix}. \quad (2)$$

We shall also write briefly $A = (a_i^{(\kappa)})$, and denote the determinant of the substitution coefficients by $|A|$. This determinant is always assumed to be nonzero. We also write the system symbolically as

$$(y_1, y_2, \dots, y_n) = A(x_1, x_2, \dots, x_n), \quad (3)$$

or, more briefly,

$$(y) = A(x). \quad (4)$$

The substitution

$$y_1 = x_1, \quad y_2 = x_2, \quad \dots, \quad y_n = x_n \quad (5)$$

or

$$J = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix} \quad (6)$$

is called the *identity substitution*.

A substitution of the form

$$y_1 = \mu_1 x_1, \quad y_2 = \mu_2 x_2, \quad \dots, \quad y_n = \mu_n x_n \quad (7)$$

or

$$M = \begin{pmatrix} \mu_1 & 0 & \cdots & 0 \\ 0 & \mu_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \mu_n \end{pmatrix} \quad (8)$$

will be called a *multiplicative substitution*, or simply a *multiplication*; the elements $\mu_1, \mu_2, \dots, \mu_n$ are called the *multipliers*.

Now suppose that we have a second linear substitution of the same dimension n , introducing a new system of variables z :

$$z_i = \sum_{h=1}^n b_h^{(i)} y_h, \quad (z) = B(y). \quad (9)$$

If we substitute for y the values from (1), then we obtain the z expressed in terms of the x by means of a new linear substitution E , which we denote by

$$z = E(x) = BA(x), \quad (10)$$

and the substitution $E = BA$ is said to be *composed* from B and A . One must, however, distinguish between AB and BA . Two substitutions A, B having the special property $AB = BA$ are called *interchangeable* or *commutative*.

The coefficients of E are obtained by inserting the expressions (1) into (9):

$$z_\kappa = \sum_h e_h^{(\kappa)} x_h, \quad e_h^{(\kappa)} = \sum_{i=1}^n b_i^{(\kappa)} a_h^{(i)}. \quad (11)$$

These formulas are exactly the same as those used earlier for the multiplication of determinants. Thus we may express the composition law of substitutions as follows.

1. To form the substitution BA composed from B and A , one proceeds exactly as if the determinants $|B|$ and $|A|$ were to be multiplied by the determinant multiplication rule. In order to obtain the elements of one row, one multiplies the elements of a row of the first factor by the corresponding elements of the columns of the second factor and then adds.

From this rule there follows immediately:

2. The determinant of a composed substitution is equal to the product of the determinants of its components.

To compose three substitutions C, B, A of the same dimension, we insert the expressions (10) for z into a further linear substitution

$$(u) = C(z). \quad (12)$$

Then the variables u are expressed in terms of the x by

$$(u) = CBA(x). \quad (13)$$

Since it is evidently immaterial whether one first introduces (10) into (12), or first expresses z by y and then y by x , one obtains for this composition the associative law

$$C(BA) = (CB)A = CBA. \quad (14)$$

The same may of course be verified directly from (11). Thus:

3. For the composition of linear substitutions, the associative law holds, but in general not the commutative law.

A multiplication in which all multipliers are equal,

$$N = \begin{pmatrix} \nu & 0 & \cdots & 0 \\ 0 & \nu & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \nu \end{pmatrix},$$

will be called a *similarity substitution*. Two substitutions that are obtained from one another by composition with a similarity substitution, such as A and AN or A and NA , are called *similar*. From the composition law one sees immediately:

4. A similarity substitution is commutative with every substitution A of the same dimension; two similarity substitutions, composed with one another, again yield a similarity substitution; and two substitutions that are each similar to a third are also similar to one another.

Equally immediate is the statement:

5. Composition with the identity substitution J leaves every substitution unchanged.

The substitution J therefore plays the role of a unit in composition and may be omitted wherever it occurs composed with other substitutions.

We have also the following theorem.

6. To every substitution A there exists one and only one inverse substitution A^{-1} satisfying

$$AA^{-1} = A^{-1}A = J. \quad (15)$$

This last statement follows from the fundamental formulas of determinant theory if one determines the elements $\alpha_h^{(\kappa)}$ of A^{-1} from the linear equations

$$\sum_i a_i^{(h)} \alpha_\kappa^{(i)} = \begin{cases} 0, & h \neq \kappa, \\ 1, & h = \kappa, \end{cases} \quad (16)$$

from which, if $A_\kappa^{(h)}$ denotes the minor of $|A|$, one obtains

$$|A| \alpha_h^{(\kappa)} = A_\kappa^{(h)}. \quad (17)$$

These in turn imply the system

$$\sum_i \alpha_i^{(h)} a_\kappa^{(i)} = \begin{cases} 0, & h \neq \kappa, \\ 1, & h = \kappa. \end{cases} \quad (18)$$

The inverse substitution of A is nothing other than the solution of the system of equations defining the direct substitution A , so that from

$$(y) = A(x)$$

there follows

$$(x) = A^{-1}(y). \quad (19)$$

For the composition of inverse substitutions one obtains from $ABB^{-1}A^{-1} = J$ the formula

$$(AB)^{-1} = B^{-1}A^{-1}. \quad (20)$$

If A, B, C are three substitutions, then composition with A^{-1} shows from either of the equations

$$AB = AC, \quad BA = CA$$

that necessarily $B = C$. Hence the characteristic properties required for a group are satisfied by the composition of substitutions, and thus:

The totality of all substitutions of a fixed dimension forms an (infinite) group.

Continuation of §37. Transformed and transposed substitutions

If a set of substitutions has the property that the composition of any two of its elements again gives an element of the same set, then that set likewise forms a group, which may be finite or infinite.

Let L be a fixed substitution. From every substitution A of the same dimension one may derive a substitution

$$A' = L^{-1}AL \quad (21)$$

called the *transform* of A by L . If we set

$$(x) = L(x'), \quad (y) = L(y'), \quad (22)$$

then from the defining relation for A it follows that

$$(y') = A'(x'), \quad (23)$$

so that passing to the transformed substitution is equivalent to applying the same change of variables L^{-1} simultaneously to both systems of variables.

If

$$A' = L^{-1}AL, \quad B' = L^{-1}BL,$$

then the composition laws give

$$A'B' = L^{-1}ABL,$$

and therefore:

7. As A runs through the substitutions of a group, the transformed substitutions $L^{-1}AL$, for fixed L , run through the substitutions of an isomorphic group.

The inverse substitutions are to be distinguished carefully from the transposed substitutions.

From each substitution A one obtains another specific substitution, called the *transposed substitution*, which for the moment we denote by A^t , by interchanging rows and columns in A - that is, by swapping the upper and lower indices of the coefficients.

If in the composition formula one interchanges lower and upper indices and at the same time exchanges the roles of a and b , one obtains the rule

$$(BA)^t = A^t B^t. \quad (24)$$

Thus the transpose behaves under composition in a way closely analogous to the inverse substitution: in taking transposes one must reverse the order of the factors, whereas for inverses one preserves the order and replaces each factor by its inverse. Combining the two rules gives

$$(A^{-1})^t = (A^t)^{-1}, \quad (25)$$

which Weber abbreviates, when no ambiguity is possible, by A_t^{-1} .

§38. The group of integral linear substitutions of one variable

We now apply the general results of the preceding paragraph to several special cases and begin with substitutions of one variable. These substitutions are identical with integral fractional-linear functions of the first degree:

$$y = \frac{\alpha x + \beta}{\gamma x + \delta}, \quad (26)$$

for which we introduce, for brevity, the symbol

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}, \quad (27)$$

and whose composition is governed by the formula

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \begin{pmatrix} \alpha' & \beta' \\ \gamma' & \delta' \end{pmatrix} = \begin{pmatrix} \alpha\alpha' + \beta\gamma' & \alpha\beta' + \beta\delta' \\ \gamma\alpha' + \delta\gamma' & \gamma\beta' + \delta\delta' \end{pmatrix}. \quad (28)$$

The coefficients $\alpha, \beta, \gamma, \delta$ are arbitrary numbers whose determinant $\alpha\delta - \beta\gamma$ is nonzero.

For the identity substitution one has

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad (29)$$

and the inverse substitution of (26) is

$$\frac{1}{D} \begin{pmatrix} \delta & -\beta \\ -\gamma & \alpha \end{pmatrix}, \quad D = \alpha\delta - \beta\gamma. \quad (30)$$

The transposed substitution attached to (26) is obtained by interchanging the diagonal entries and simultaneously changing the signs of the off-diagonal entries:

$$\begin{pmatrix} \delta & -\beta \\ -\gamma & \alpha \end{pmatrix}, \quad (31)$$

or, what is the same, by replacing $\alpha, \beta, \gamma, \delta$ by $\delta, -\gamma, -\beta, \alpha$. One verifies immediately that

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \begin{pmatrix} \delta & -\beta \\ -\gamma & \alpha \end{pmatrix} = \begin{pmatrix} D & 0 \\ 0 & D \end{pmatrix}, \quad (32)$$

which on the one hand confirms formula (24), and on the other shows that every such substitution commutes with its transpose whenever D is a symmetric function of α and δ .

§39. Permutation groups as linear substitution groups

Let $x_1, x_2, \dots, x_n$ be a system of n variables, and let $\alpha_1, \alpha_2, \dots, \alpha_n$ be some ordering of the digits $1, 2, \dots, n$. Then the equations

$$x_1 = x'_{\alpha_1}, \quad x_2 = x'_{\alpha_2}, \quad \dots, \quad x_n = x'_{\alpha_n} \quad (33)$$

determine a linear substitution of dimension n ,

$$A = (a_i^{(k)}), \quad (x) = A(x'), \quad (34)$$

in which each row and each column contains exactly one nonzero coefficient, and that coefficient is equal to 1.

The determinant $|A|$ is therefore equal to ± 1 . If, after carrying out the substitution, one sets $x'_i = x_i$ again, the result is simply the permutation

$$\begin{pmatrix} 1, 2, \dots, n \\ \alpha_1, \alpha_2, \dots, \alpha_n \end{pmatrix}$$

of the indices of the variables.

If B is a second substitution of the same kind,

$$(x') = B(x''), \quad (35)$$

or explicitly

$$x'_1 = x''_{\beta_1}, \quad x'_2 = x''_{\beta_2}, \quad \dots, \quad x'_n = x''_{\beta_n}, \quad (36)$$

then the composition rules of §37 yield

$$x_1 = x''_{\beta\alpha_1}, \quad x_2 = x''_{\beta\alpha_2}, \quad \dots, \quad x_n = x''_{\beta\alpha_n}, \quad (37)$$

which is abbreviated by

$$(x) = AB(x''). \quad (38)$$

But according to the composition law for permutations one has

$$\begin{pmatrix} 1, 2, \dots, n \\ \alpha_1, \alpha_2, \dots, \alpha_n \end{pmatrix} \begin{pmatrix} 1, 2, \dots, n \\ \beta_1, \beta_2, \dots, \beta_n \end{pmatrix} = \begin{pmatrix} 1, 2, \dots, n \\ \beta\alpha_1, \beta\alpha_2, \dots, \beta\alpha_n \end{pmatrix}. \quad (39)$$

Thus, if the substitutions A and B are regarded as permutations of the indices, then the substitution AB is exactly the composed permutation AB .

Hence permutation groups on n symbols are nothing other than a special case of finite groups of linear substitutions of dimension n .

Permutations of the first kind correspond to substitutions with determinant $+1$, and permutations of the second kind correspond to substitutions with determinant -1 .

This relation between permutations and linear substitutions enables us to derive from every permutation group a group of linear substitutions, and conversely to settle certain questions about permutation groups with the aid of the theory of linear substitutions.

§40. Groups of binary linear substitutions with real coefficients

For the binary substitutions

$$y_1 = \alpha x_1 + \beta x_2, \quad y_2 = \gamma x_1 + \delta x_2, \quad (1)$$

the determinant is

$$D = \alpha\delta - \beta\gamma.$$

We shall first suppose that the coefficients $\alpha, \beta, \gamma, \delta$ are real, and ask under what conditions a group of such substitutions can be finite.

The characteristic equation of the substitution is

$$\begin{vmatrix} \alpha - \lambda & \beta \\ \gamma & \delta - \lambda \end{vmatrix} = \lambda^2 - (\alpha + \delta)\lambda + D = 0. \quad (2)$$

Its roots are

$$\lambda = \frac{\alpha + \delta \pm \sqrt{(\alpha + \delta)^2 - 4D}}{2}. \quad (3)$$

If

$$(\alpha + \delta)^2 - 4D > 0,$$

then the roots are real and distinct. In that case the substitution can be brought, by a similar substitution, into the form

$$y_1 = \lambda_1 x_1, \quad y_2 = \lambda_2 x_2. \quad (4)$$

The n -fold iterate is then

$$y_1 = \lambda_1^n x_1, \quad y_2 = \lambda_2^n x_2.$$

For some positive power to become the identity, λ_1 and λ_2 must be roots of unity. Since they are real, this forces

$$\lambda_1, \lambda_2 \in \{1, -1\}.$$

If

$$(\alpha + \delta)^2 - 4D = 0,$$

then the two characteristic roots coincide: $\lambda_1 = \lambda_2 = \lambda$. The substitution can then either be reduced to the diagonal form above, or to the Jordan form

$$y_1 = \lambda x_1, \quad y_2 = x_1 + \lambda x_2. \quad (5)$$

In the latter case the n -fold iterate contains the term

$$y_2 = n\lambda^{n-1}x_1 + \lambda^n x_2,$$

and therefore cannot be the identity for any positive n , unless the nilpotent part vanishes. Thus a finite group contains no nontrivial substitution of this Jordan type.

It remains to consider the case

$$(\alpha + \delta)^2 - 4D < 0.$$

Then the two characteristic roots are conjugate complex roots. A real substitution with such roots is, over the real field, similar to a rotation-dilatation. If the substitution has finite order, its characteristic roots must again be roots of unity; hence the quotient of the two roots is a root of unity and the corresponding projective transformation is periodic.

Consequently, a real binary linear substitution which belongs to a finite group is characterized by the fact that, after multiplication by a scalar if necessary, it can be interpreted as a rotation through an angle commensurable with 2π , or as one of the exceptional involutory cases in which the characteristic roots are 1 and -1 .

This gives a first reduction of the problem of finite real binary substitution groups to the problem of finite rotation groups and their associated projective actions.

§41. Invariants and covariants of groups of linear substitutions

The theory of linear substitutions is connected in the closest way with the theory of algebraic forms. By a *form* in the variables

$$x_1, x_2, \dots, x_n$$

we mean an integral homogeneous function of these variables. A form

$$f(x_1, \dots, x_n)$$

of degree p is transformed by a linear substitution A into another form

$$f'(x'_1, \dots, x'_n)$$

of the same degree in the new variables. If $f' = f$, that is, if the form is transformed into itself by the substitution, then f is called an *invariant* of A .

More generally, suppose that f is a function not only of the variables x , but also of other variables $u, v, \dots$. If f is transformed into itself by applying A to the x -variables and at the same time applying corresponding substitutions to the variables $u, v, \dots$, then f is called a *covariant*.

We now consider a group S of linear substitutions in the variables

$$x_1, \dots, x_n,$$

and seek those forms which are simultaneously invariants, or covariants, for all substitutions of the group.

Let

$$F_1, F_2, \dots, F_r$$

be a system of such invariants which are rationally independent, meaning that no identical rational relation holds among them, while every other invariant can be rationally expressed through them. Such a system is called a *basis* of the invariant system of the group S .

The most general invariant has the form

$$F = A_1 F_1 + A_2 F_2 + \dots + A_r F_r, \tag{6}$$

where $A_1, A_2, \dots, A_r$ are arbitrary constants, or more generally quantities lying in the appropriate coefficient field.

From the general theory of algebraic forms one obtains the following fundamental facts.

- I. There always exists a finite system of invariants from which all the remaining invariants may be generated rationally.
- II. If the group is finite, the invariant system may be chosen so that it reflects the finite averaging process over the substitutions of the group.
- III. The corresponding covariants are obtained in the same manner once the simultaneous transformation law of the auxiliary variables is fixed.

These facts allow the invariant theory of a finite linear substitution group to be treated algebraically: the full system of invariant forms is replaced by a finite basis and the rational relations among its elements.

§42. The absolute invariant and the ground field

Let

$$F_1, F_2, \dots, F_r$$

be a basis of the invariant system J , as determined above, and let F be any other nonconstant invariant of S . The forms $A_1, A_2, \dots, A_r$ may then be chosen so that

$$F = A_1 F_1 + A_2 F_2 + \dots + A_r F_r. \quad (7)$$

Apply all substitutions of the group S to this identity. By assumption,

$$F, F_1, F_2, \dots, F_r$$

remain unchanged; the coefficients $A_1, A_2, A_3, \dots$, however, may be transformed into

$$A_1^{(i)}, A_2^{(i)}, A_3^{(i)}, \dots$$

If one adds all identities obtained in this way and divides by the order s of the group S , then one obtains

$$F = \bar{A}_1 F_1 + \bar{A}_2 F_2 + \dots + \bar{A}_r F_r, \quad (8)$$

where now $\bar{A}_1, \bar{A}_2, \dots$ are themselves invariants of the group S . By the defining property of the basis, these averaged coefficients are rationally expressible through $F_1, \dots, F_r$.

Hence:

Theorem. *Every invariant F can be represented as a rational function of the basis invariants $F_1, \dots, F_r$.*

Let

$$\vartheta_1, \vartheta_2, \dots, \vartheta_h$$

be the coefficients of the general substitution of S , and let Ω be the rational field generated by these quantities. Every invariant F is then a function of the variables x and of the quantities ϑ . If the x 's are treated as independent variables and the ϑ 's as parameters, the totality of all invariants so obtained forms a field. We shall call it the *field of invariants* of the group S .

The absolute invariants are precisely those functions that remain unchanged under all substitutions of the group and depend only on the orbit of the point under the group action. Thus they furnish the rational functions on the quotient of the space of variables by the group.

§43. The form problem

The problem of determining the values of

$$x_1, x_2, \dots, x_n$$

which give a prescribed value F_0 to a given invariant F , when the coefficients of the group are assumed known, is called the *form problem* of the group S .

Let $\Omega(t)$ denote a general function of the field Ω which is carried into itself by the substitutions of S . The roots of the equation

$$\Omega(t) = \omega,$$

where ω is any fixed value, are all distinct and are transformed into one another by the substitutions of S .

The solution of the form problem is closely connected with the solution of the equation

$$\Omega(t) = \omega.$$

Indeed, if ϑ is a root of this equation and one sets

$$x_i = \vartheta_i(\vartheta), \tag{9}$$

for suitable functions ϑ_i , then one obtains values of the x_i satisfying all invariant relations.

If the invariant J can be represented as a rational function of ϑ , then this function cannot change when any of the substitutions

$$(\vartheta, \vartheta_1), \quad (\vartheta, \vartheta_2), \quad \dots$$

is applied. Consequently it is rationally expressible through the coefficients of $\Omega(t)$. Whether this representation is possible depends on the nature of the group.

If S' is a subgroup of S of index j , and no larger subgroup has the same invariance property, then ϑ satisfies an equation of degree j . This equation is to be regarded as a *resolvent* of the form problem. Every function which admits exactly the substitutions of S' satisfies such a resolvent.

Thus the form problem transforms invariant theory into an equation problem: the passage from a group to its subgroups corresponds to the passage from a general equation to its resolvents.

§44. Galois groups and resolvents

The theory of groups of linear substitutions is closely related to the theory of algebraic equations. Let

$$t^n + a_1 t^{n-1} + a_2 t^{n-2} + \cdots + a_n = 0 \quad (10)$$

be an equation of degree n , with roots

$$\alpha_1, \alpha_2, \dots, \alpha_n.$$

The totality of all permutations of these roots which leave unchanged every rational relation among them forms the *Galois group* of the equation.

A function

$$F(\alpha_1, \dots, \alpha_n)$$

of the roots that is unchanged by all permutations of the Galois group is rationally expressible through the coefficients

$$a_1, \dots, a_n$$

of the equation.

Solving the equation is equivalent to determining a function ϑ of the roots that is left fixed by no nonidentity permutation of the Galois group. Such a function is called a *primitive function* of the field.

The Galois group may always be regarded as a group of linear substitutions: replace each permutation of the roots by the corresponding substitution of the variables

$$x_1, \dots, x_n,$$

as described in §39. In this way the group of the equation becomes a concrete linear substitution group, and the methods of invariant theory become available for the study of algebraic equations.

Resolvents arise by passing to subgroups. If a function of the roots is invariant under a subgroup S' but not under the whole group S , then the conjugates of that function under S are indexed by the cosets of S' . Their elementary symmetric functions are invariant under S , hence rational functions of the coefficients of the original equation. The function therefore satisfies an auxiliary equation whose degree is the index $[S : S']$.

§45. Metacyclic groups

A group S is called *metacyclic* if it has a distinguished chain of subgroups

$$S, S_1, S_2, \dots, S_\nu = 1, \quad (11)$$

such that each S_i is a normal subgroup of S_{i-1} and each quotient group

$$S_{i-1}/S_i$$

is cyclic.

Such a group can be built by solving a chain of equations, each of which is cyclic. The solvability of an algebraic equation by radicals is therefore expressed group-theoretically by the metacyclic nature of its Galois group, or equivalently by the existence of a suitable chain with cyclic factors.

For the symmetric group on n letters, metacyclicity occurs only for $n \leq 4$. Likewise the alternating group on n letters is metacyclic only for $n \leq 4$. This is the group-theoretic obstruction behind the impossibility, in general, of solving equations of degree at least five by radicals.

§46. Abstract and concrete groups

Until now the exposition of group theory has been tied to definite operations: substitutions, linear transformations, and similar concrete processes. But the group concept may also be detached from any particular representation and defined purely formally.

By an *abstract group* one understands a system of elements

$$A, B, C, \dots$$

for which a composition law, or multiplication, is defined and satisfies the following rules.

1. From any two elements A, B , their product is a uniquely determined element $C = AB$.
2. The associative law holds:

$$(AB)C = A(BC).$$

3. There is an *identity element* J such that

$$JA = AJ = A$$

for every element A .

4. For every element A there is an inverse element A^{-1} such that

$$AA^{-1} = A^{-1}A = J.$$

Two abstract groups are called *isomorphic* if their elements can be named in such a way that the multiplication rule is the same in both groups. This definition separates the intrinsic structure of the group from the particular objects on which the group happens to act.

The point of abstraction is not to discard concrete examples, but to recognize when two different systems of operations have the same internal law of composition. Permutation groups, linear substitution groups, rotation groups, and Galois groups can then be compared by their multiplication tables rather than by their external form.

§47. The alternating group and its invariants

The alternating group on n letters consists of all even permutations and has index 2 in the symmetric group. The function

$$\Delta = \prod_{i < k} (x_i - x_k) \quad (12)$$

is an invariant of the alternating group. More precisely, it is an invariant which changes sign under every odd permutation.

The square

$$\Delta^2$$

is therefore an absolute invariant of the symmetric group. It is a symmetric function of the x_i , and is represented by the discriminant of the corresponding equation.

The most general invariant of the alternating group can be written in the form

$$F = F_1 + \Delta F_2, \quad (13)$$

where F_1 and F_2 are symmetric functions, that is, invariants of the full symmetric group.

For $n \geq 5$, the alternating group is simple: it has no normal subgroup other than the identity and itself. This fact is one of the central structural reasons why the general equation of degree at least five cannot be solved by radicals.

§48. Orthogonal substitutions

A linear substitution is called *orthogonal* if it transforms the quadratic form

$$x_1^2 + x_2^2 + \cdots + x_n^2 \quad (14)$$

into itself. For a substitution A with coefficients $a_i^{(k)}$, this means that the identity

$$\sum_k \left(\sum_i a_i^{(k)} x_i \right)^2 = \sum_i x_i^2 \quad (15)$$

holds in the variables x .

Comparing coefficients gives the relations

$$\sum_k a_i^{(k)} a_j^{(k)} = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases} \quad (16)$$

For example, if

$$A = \begin{pmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{pmatrix},$$

then the orthogonality relations are

$$a_1^2 + b_1^2 + c_1^2 = 1, \quad a_1 a_2 + b_1 b_2 + c_1 c_2 = 0, \quad (17)$$

$$a_2^2 + b_2^2 + c_2^2 = 1, \quad a_1 a_3 + b_1 b_3 + c_1 c_3 = 0, \quad (18)$$

$$a_3^2 + b_3^2 + c_3^2 = 1, \quad a_2 a_3 + b_2 b_3 + c_2 c_3 = 0. \quad (19)$$

These equations say exactly that the row vectors of A form an orthonormal system.

From these relations one obtains

$$|A|^2 = 1,$$

and hence

$$|A| = \pm 1.$$

The orthogonal substitutions with determinant $+1$ are those of the *first kind*, or rotations. Those with determinant -1 are those of the *second kind*, or reflections.

The totality of all orthogonal substitutions in n variables forms a group, the *orthogonal group*. One may regard either the positions of the body or the rotations themselves as the elements of the group.

§49. Linear fractional substitutions

We shall now show that the group of ternary orthogonal substitutions is isomorphic to the group of linear fractional substitutions in one variable, or equivalently to the group of binary substitutions acting on ratios.

Write a ternary orthogonal substitution as

$$(y_1, y_2, y_3) = A(x_1, x_2, x_3), \quad (20)$$

so that

$$y_1^2 + y_2^2 + y_3^2 = x_1^2 + x_2^2 + x_3^2. \quad (21)$$

We first apply a fixed, nonorthogonal change of variables of the form

$$x_1 = \xi_1 + \xi_2, \quad y_1 = \eta_1 + \eta_2, \quad (22)$$

$$x_2 = \xi_1 - \xi_2, \quad y_2 = \eta_1 - \eta_2, \quad (23)$$

$$x_3 = \xi_3, \quad y_3 = \eta_3. \quad (24)$$

Under this transformation the quadratic condition is converted into an equivalent condition in the new variables.

Now introduce binary variables by setting, in effect,

$$x_1 = \sqrt{2} \xi_1 \xi_3, \quad x_2 = \sqrt{2} \xi_2 \xi_3.$$

Then the corresponding quadratic relation vanishes identically. The transformed variables y_1, y_2, y_3 become binary quadratic forms in ξ_1, ξ_2 , and these forms must satisfy the transformed quadratic equation identically.

The product relation then forces the relevant quadratic forms to decompose into linear factors. Since y_2 and y_3 cannot have a common factor without making the determinant of the transformation vanish, the factors are arranged in such a way that the ternary orthogonal substitution is represented by a binary linear substitution, considered projectively.

Thus ternary rotations are represented by fractional linear transformations. In modern language, this is the classical passage from the binary linear group, modulo scalar multiplication, to the rotation group in three dimensions.

§51. Finite groups of linear fractional substitutions; poles of groups

From the preceding developments it follows that, once all finite groups of linear fractional substitutions are known, one immediately obtains all finite groups of proper ternary orthogonal substitutions and all finite groups of binary linear substitutions with determinant 1.

Let S be a finite group of linear fractional substitutions in a variable x . Write its substitutions in canonical form

$$x' = \frac{\alpha x + \beta}{\gamma x + \delta}, \quad (25)$$

with determinant condition

$$\alpha\delta - \beta\gamma = 1.$$

A nonidentity substitution of this form generally has two fixed points, or *poles*, obtained from the quadratic equation

$$\gamma x^2 + (\delta - \alpha)x - \beta = 0. \quad (26)$$

Only in the exceptional case $\gamma = 0$ and $\alpha = \delta$, that is, for a translation $x' = x + \beta$, does the equation reduce to degree one; then there is only one improper pole, situated at infinity.

The pole structure is the key to the classification of finite fractional linear groups. A finite group acts on the sphere of projective values, and the exceptional points with nontrivial stabilizer determine the group.

§52. The cyclic and dihedral groups

The simplest finite groups are the *cyclic groups*. A cyclic group of degree n is generated by the substitution

$$x' = \varepsilon x, \quad (27)$$

where ε is a primitive n -th root of unity. The n substitutions of the group are

$$x, \varepsilon x, \varepsilon^2 x, \dots, \varepsilon^{n-1} x. \quad (28)$$

The two poles are 0 and ∞ , and both are fixed by every substitution of the group.

A *dihedral group* of degree $2n$ contains, besides these substitutions, the n substitutions

$$x' = \frac{\varepsilon^k}{x}, \quad k = 0, 1, \dots, n-1. \quad (29)$$

These interchange the two poles 0 and ∞ . Thus the group consists of the substitutions

$$x' = \varepsilon^k x, \quad x' = \frac{\varepsilon^k}{x}, \quad k = 0, 1, \dots, n-1. \quad (30)$$

The element ε must run through a full system of n -th roots of unity. Hence, if the ε occurring in the generator is primitive, the whole cyclic subgroup is obtained from its powers, and the dihedral group is obtained by adjoining the inversion $x \mapsto 1/x$, together with its products with the rotations.

§58. The Icosahedral Group

Since in the icosahedral group there is only one system of conjugate fivefold poles, we may, by the preceding reduction, assume that the group contains the two substitutions

$$\Theta(x) = \varepsilon x, \quad \psi(x) = -\frac{1}{x}, \quad (31)$$

where ε is a primitive fifth root of unity. We choose here, as we are free to do, the second substitution in the form $-1 : x$, because the formulae thereby become simpler.

The basic form of degree twelve belonging to the system of fivefold poles must, since it admits the substitutions (??), have the form

$$f(x_1, x_2) = x_1 x_2 (x_1^{10} + m x_1^5 x_2^5 - x_2^{10}), \quad (32)$$

and it remains to determine the constant factor m . The two other basic forms have degrees twenty and thirty. We can determine m from the theorem of §54, if we can construct from f a covariant of degree less than sixty which cannot be represented as a product of powers of three forms of degrees twelve, twenty, and thirty. Such a covariant must then vanish identically.

A covariant of degree sixteen of the form f is easily obtained by taking, in the sense of the first volume, the fourth polar of the binary form $f(x_1, x_2)$:

$$12 \cdot 11 \cdot 10 \cdot 9 P_4(x, \xi) = u_0 \xi_1^4 + 4u_1 \xi_1^3 \xi_2 + 6u_2 \xi_1^2 \xi_2^2 + 4u_3 \xi_1 \xi_2^3 + u_4 \xi_2^4, \quad (33)$$

where the u_i are obtained from the fourth partial derivatives of f . From this fourth polar one forms, as in the theory of binary quartics, its first invariant,

$$u_0 u_4 - 4u_1 u_3 + 3u_2^2. \quad (34)$$

Since a form of degree sixteen cannot be a product of forms of degrees twelve, twenty, and thirty, this covariant must vanish identically.

For the form (??) one has, up to the common numerical factor coming from the fourth differentiation,

$$\begin{aligned} u_0 &= 11 \cdot 10 \cdot 9 \cdot 8 x_1^7 x_2 + 6 \cdot 5 \cdot 4 \cdot 3 m x_1^2 x_2^6, \\ u_1 &= 4 \cdot 11 \cdot 10 \cdot 9 x_1^6 x_2^2 + 4 \cdot 6 \cdot 5 \cdot 4 m x_1 x_2^7, \\ u_2 &= 6 \cdot 5 \cdot m x_1^5 x_2^5, \\ u_3 &= -4 \cdot 11 \cdot 10 \cdot 9 x_1^2 x_2^6 + 4 \cdot 6 \cdot 5 \cdot 4 m x_1^7 x_2, \\ u_4 &= -11 \cdot 10 \cdot 9 \cdot 8 x_1 x_2^7 + 6 \cdot 5 \cdot 4 \cdot 3 m x_1^6 x_2^2. \end{aligned}$$

Substitution in (??), and comparison of the coefficient of the appropriate middle term, gives

$$m = \pm 11.$$

Both signs are admissible. The one choice is carried into the other by replacing x_1 by $-x_1$, or equivalently by an allowable transformation of the group. We therefore put

$$f(x_1, x_2) = x_1 x_2 (x_1^{10} + 11 x_1^5 x_2^5 - x_2^{10}). \quad (35)$$

The ten remaining fivefold poles are consequently obtained by solving

$$x^{10} + 11x^5 - 1 = 0.$$

Writing

$$\omega = -\frac{11 + 5\sqrt{5}}{2}, \quad \omega' = -\frac{11 - 5\sqrt{5}}{2}, \quad (36)$$

so that

$$\omega^2 + \omega = 1, \quad \omega + \omega' = -1, \quad \omega\omega' = -1, \quad (37)$$

these fivefold poles, besides 0 and ∞ , may be written in the form

$$x = \varepsilon^\lambda \omega, \quad x = \varepsilon^\lambda \omega', \quad \lambda = 0, 1, 2, 3, 4. \quad (38)$$

Here $\varepsilon + \varepsilon^{-1} = 2 \cos(2\pi/5) = (-1 + \sqrt{5})/2$ and $\varepsilon^2 + \varepsilon^{-2} = 2 \cos(4\pi/5) = (-1 - \sqrt{5})/2$, so that the notation is compatible with the elementary identities just displayed.

We now apply the theorem of §53. If a, a' are the two poles of a substitution Θ , and if b is a pole conjugate with a , then there is a substitution χ of the group such that $b = \chi(a)$, and such that $b' = \chi(a')$ is the second pole of $\chi^{-1}\Theta\chi$. We may take $a = \infty$, $a' = 0$, and $b = \omega$. Then b' is a still undetermined pole ξ among those in (??), and χ has the form

$$\chi(x) = \frac{\omega\alpha x + \beta}{\alpha x + \beta\omega^{-1}}, \quad \alpha\beta(\omega^{-1} - \omega) = 1. \quad (39)$$

For every exponent h not divisible by five, the substitution Θ^h may be used. The poles of $\chi^{-1}\Theta^h\chi$ must be ω and ξ . From the corresponding quadratic equation for the fixed points one obtains

$$\frac{\beta}{\alpha} = \pm\omega\sqrt{-\xi\omega^{-1}},$$

and the condition (??) then determines the remaining scalar factor. In Weber's normalization this reduces to the simple form

$$\chi(x) = \frac{\omega x + \beta}{x - \omega^{-1}}, \quad (40)$$

where β is still to be determined and, from the determinant condition, has the value $-\omega\xi$.

If χ is known, the whole icosahedral group can be generated. Let r, s run through the residues 0, 1, 2, 3, 4. The group then contains the substitutions

$$\Theta^r, \quad \psi\Theta^r, \quad \Theta^r\chi\Theta^s, \quad \Theta^r\chi\psi\Theta^s. \quad (41)$$

There are sixty of them. Written as matrices of fractional-linear substitutions, they are

$$\Theta^r = \begin{pmatrix} \varepsilon^r & 0 \\ 0 & 1 \end{pmatrix}, \quad (42)$$

$$\psi\Theta^r = \begin{pmatrix} 0 & 1 \\ -\varepsilon^r & 0 \end{pmatrix}, \quad (43)$$

$$\Theta^r\chi\Theta^s = \begin{pmatrix} \varepsilon^r\omega & \varepsilon^{r-s}\beta \\ 1 & -\varepsilon^{-s}\omega \end{pmatrix}, \quad (44)$$

$$\Theta^r\chi\psi\Theta^s = \begin{pmatrix} -\varepsilon^{r+s}\beta & \varepsilon^r\omega \\ \varepsilon^s\omega & 1 \end{pmatrix} = \begin{pmatrix} -\varepsilon^r\beta\omega^{-1} & \varepsilon^{r-s} \\ 1 & \varepsilon^{-s}\omega^{-1} \end{pmatrix}. \quad (45)$$

Since $-\omega^2$ is no power of ε , no two of these matrices represent the same substitution.

The icosahedral group has, as we have seen, twelve fivefold poles. Any two of these are the poles of four substitutions of fifth order; together with the identity these form a cycle of five elements. Thus there are twenty-four elements of fifth order. Similarly, the twenty threefold poles lead to twenty substitutions of third order, and the thirty twofold poles, paired two by two, lead to fifteen substitutions of second order. Together with the identity this gives

$$24 + 20 + 15 + 1 = 60.$$

The determination of β in (??) is obtained by considering the orders of the substitutions (??) and (??). For an arbitrary linear substitution

$$S = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

the second and third iterates are

$$S^2 = \begin{pmatrix} a^2 + bc & b(a + d) \\ c(a + d) & bc + d^2 \end{pmatrix},$$

$$S^3 = \begin{pmatrix} a^3 + 2abc + bcd & b(a^2 + bc + ad + d^2) \\ c(a^2 + bc + ad + d^2) & d^3 + abc + 2bcd \end{pmatrix}.$$

Disregarding the cases where b or c vanishes, which do not occur here, one obtains the necessary and sufficient conditions

$$a + d = 0 \tag{46}$$

for a substitution of second order, and

$$a^2 + bc + ad + d^2 = 0 \tag{47}$$

for one of third order.

The substitutions (??), apart from the identity, have fifth order. The five substitutions (??) have second order. Among the substitutions (??), exactly the five of the form $\Theta^r \chi \Theta^{-r}$ have second order. Hence five among the substitutions (??) must also have second order. From (??) this is possible only if β is a power of ε .

The substitutions of third order must now be distributed among the forms (??) and (??). The condition that (??) have third order is

$$\omega^2(\varepsilon^r + \varepsilon^{-s} + \varepsilon^{-(r+s)}) = 1, \tag{48}$$

and for (??) the condition is

$$\omega^2 = \beta - \varepsilon^{-(r+s)} - \beta^2 \varepsilon^{r+s}. \tag{49}$$

Either one of these conditions can be satisfied only when

$$\beta = 1.$$

Indeed, in (??) the left side is real only when β is real; since β is a power of ε , this forces $\beta = 1$. In (??) one obtains the same conclusion by passing to conjugate-imaginary quantities; the expression can remain invariant under conjugation only if $\beta = \beta^{-1}$, hence $\beta = 1$ among fifth roots of unity.

With $\beta = 1$, five of the substitutions (??), namely those of the form $\Theta^r \chi \psi \Theta^{-r}$, have second order. The substitutions (??) have third order precisely when

$$r + s \equiv \pm 2 \pmod{5}, \tag{50}$$

and the substitutions (??) have third order precisely when

$$r + s \equiv \pm 1 \pmod{5}. \tag{51}$$

Thus exactly ten substitutions of second order and twenty of third order occur in the forms (??) and (??); the remaining cases are of fifth order.

Summarizing, χ must have the form

$$\chi(x) = \frac{\omega x + 1}{x - \omega^{-1}}. \tag{52}$$

Among the sixty substitutions

$$\Theta^r, \quad \psi\Theta^r, \quad \Theta^r\chi\Theta^s, \quad \Theta^r\chi\psi\Theta^s, \quad (53)$$

there occur, besides the identity,

1. fifteen substitutions of second order:

$$\psi\Theta^r, \quad \Theta^r\chi\Theta^{-r}, \quad \Theta^r\chi\psi\Theta^{-r} \quad (r = 0, 1, 2, 3, 4);$$

2. twenty substitutions of third order:

$$\Theta^r\chi\Theta^s \quad (r + s \equiv \pm 2), \quad \Theta^r\chi\psi\Theta^s \quad (r + s \equiv \pm 1);$$

3. twenty-four substitutions of fifth order:

$$\Theta^r \quad (r \neq 0), \quad \Theta^r\chi\Theta^s \quad (r + s \equiv \pm 1), \quad \Theta^r\chi\psi\Theta^s \quad (r + s \equiv \pm 2).$$

In explicit fractional-linear form these substitutions include

$$\varepsilon^r x, \quad -\frac{1}{\varepsilon^r x}, \quad \frac{\varepsilon^r \omega x + \varepsilon^{r-s}}{x - \varepsilon^{-s} \omega}, \quad \frac{-\varepsilon^r \omega^{-1} x + \varepsilon^{r-s}}{x + \varepsilon^{-s} \omega^{-1}}. \quad (54)$$

It remains only to verify that the totality of substitutions (??) is indeed a group. This follows from the composition laws, which we now record. First, directly from the meaning of Θ, ψ, χ ,

$$\Theta^5 = 1, \quad \psi^2 = 1, \quad \psi\Theta = \Theta^{-1}\psi, \quad \chi\psi = \psi\chi. \quad (55)$$

For an arbitrary exponent λ , use

$$\omega = \varepsilon + \varepsilon^{-1}, \quad \omega' = \varepsilon^2 + \varepsilon^{-2}, \quad \omega\omega' = -1. \quad (56)$$

Then, putting first $\lambda = 1$ and then $\lambda = 2$, Weber obtains

$$\chi\Theta^{\pm 1}\chi = \Theta^{\pm 1}\chi\psi\Theta^{\pm 1}, \quad \chi\Theta^{\pm 2}\chi = \Theta^{\pm 2}\chi\Theta^{\pm 2}. \quad (57)$$

Together with (??), these formulae allow any product of two substitutions of the form (??) to be reduced again to one of the same forms. Thus the group property is proved.

From the first formula in (??) one also derives

$$\psi = \Theta\chi^{-1}\Theta^{-1}\chi^{-1}\Theta,$$

so that ψ may be derived from Θ and χ . The substitutions Θ and χ alone may therefore be taken as generators of the icosahedral group.

To write χ with determinant one, we use

$$-\omega^{-2} - 1 = -\omega^{-1} = (\varepsilon - \varepsilon^{-1})^2.$$

Replacing ω by $\varepsilon + \varepsilon^{-1}$, this gives

$$\chi = \frac{1}{\varepsilon - \varepsilon^{-1}} \begin{pmatrix} \omega & 1 \\ 1 & -\omega^{-1} \end{pmatrix} = \begin{pmatrix} \frac{\omega}{\varepsilon - \varepsilon^{-1}} & \frac{1}{\varepsilon - \varepsilon^{-1}} \\ \frac{1}{\varepsilon - \varepsilon^{-1}} & -\frac{\omega^{-1}}{\varepsilon - \varepsilon^{-1}} \end{pmatrix}. \quad (58)$$

§59. Divisors of the Icosahedral Group

The divisors, or subgroups, of the icosahedral group must be sought among the lower polyhedral groups. First there are cyclic groups C_n , each generated by one of the icosahedral substitutions. Counting them immediately from the preceding arrangement according to the orders of the substitutions gives

$$15 \text{ groups } C_2, \quad 10 \text{ groups } C_3, \quad 6 \text{ groups } C_5.$$

There are also dihedral groups D_2, D_3, D_5 in the icosahedral group. Representatives may be written in the form

$$\begin{aligned} D_2 : & 1, \chi\psi, \Theta^{-1}\chi\psi\Theta^{-1}, \Theta\chi\psi\Theta, \\ D_3 : & \Theta^r\chi\Theta^s \quad (r+s \equiv \pm 2 \pmod{5}), \\ D_5 : & \Theta^r, \psi\Theta^r, \quad r = 0, 1, 2, 3, 4. \end{aligned}$$

The number of such dihedral groups is found from the fact that a dihedral group is completely determined by its cyclic subgroup. For the four-group D_2 one must remember that the same group is obtained whether one begins with ψ , $\chi\psi$, or $\Theta\chi\psi\Theta$; hence the directly obtained number is to be divided by three. Thus the numbers of groups D_2, D_3, D_5 are respectively

$$5, \quad 10, \quad 6.$$

Of special importance are the tetrahedral groups contained in the icosahedral group. We determine one of them, denoted by Q . A tetrahedral group must contain a four-group D_2 ; we take the one generated by ψ and $\chi\psi$. In addition, Q must contain a substitution of third order. Such a substitution can be taken in one of the two forms

$$\Theta^r\chi\Theta^s, \quad \Theta^r\chi\psi\Theta^s.$$

Both choices lead to the same result; choose

$$\varphi = \Theta^r\chi\Theta^s, \quad r+s \equiv \pm 2 \pmod{5}. \quad (59)$$

Neither of the numbers r, s may be zero; otherwise either $\chi\varphi$ or $\varphi\chi$ would be a power of Θ , hence of fifth order, while no element of fifth order can occur in Q .

Using the composition laws of §58 one obtains

$$\chi\varphi = \Theta^r\chi\psi\Theta^{r+s} \quad (r \neq \pm 1), \quad \chi\varphi = \Theta^{-r}\chi\Theta^{-r+s} \quad (r \neq \pm 2),$$

with the exclusions required by the same reason just given. The four possible cases left are

$$r = \pm 1, \quad s = \pm 2, \quad \text{or} \quad r = \pm 2, \quad s = \pm 1 \pmod{5}.$$

If one of the four corresponding substitutions of third order belongs to Q , the other three do also. Put

$$\varphi = \Theta\chi\Theta. \quad (60)$$

Then from the relations of §58 one obtains

$$\varphi^2 = \Theta^{-1}\chi\Theta^{-1}, \quad \varphi\chi = \Theta^{-1}\chi\psi\Theta^{-2}, \quad \chi\varphi = \Theta^2\chi\psi\Theta, \quad (61)$$

and, by adjoining the compositions with ψ , one sees that the same group is obtained if one begins with a substitution of the second form.

The whole group Q is therefore completely determined by the chosen four-group D_2 . It contains the elements

$$1, \quad \psi, \quad \chi\psi, \quad \Theta^{-1}\chi\psi\Theta^{-1}, \quad \varphi, \quad \varphi\psi, \quad \varphi\chi, \quad \varphi\chi\psi, \quad \varphi^2, \quad \varphi^2\psi, \quad \varphi^2\chi, \quad \varphi^2\chi\psi. \quad (62)$$

Equivalently, Q is generated by φ, ψ, χ and is represented in the form of a tetrahedral group. The multiplication rules following from (??) are

$$\psi\varphi^{-1} = \chi\varphi^{-1}\psi, \quad \chi\varphi = \varphi\chi\psi, \quad \psi\chi = \chi\psi, \quad \chi^2 = \psi^2 = 1. \quad (63)$$

They are the defining composition laws of the tetrahedral group.

Since the tetrahedral group contains five four-groups D_2 , the icosahedral group contains five tetrahedral subgroups. They are obtained from Q by transforming with the powers of Θ :

$$\Theta^{-r}Q\Theta^r, \quad r = 0, 1, 2, 3, 4.$$

No two of these groups have, besides the identity, a common substitution. Indeed, Q and $\Theta^{-1}Q\Theta$ have in common, besides the identity, only two displayed elements, and neither of these occurs in $Q\Theta^{-1}$.

It follows from the theorem on permutation representations by cosets that the icosahedral group is isomorphic to a permutation group of degree sixty on five letters. The representation is obtained by adjoining to the five right cosets

$$Q, \quad Q\Theta, \quad Q\Theta^2, \quad Q\Theta^3, \quad Q\Theta^4$$

each element δ of the icosahedral group, forming

$$Q\delta, \quad Q\Theta\delta, \quad Q\Theta^2\delta, \quad Q\Theta^3\delta, \quad Q\Theta^4\delta,$$

and examining the permutation so produced. For the generating substitutions $\delta = \Theta, \chi$, one obtains respectively the permutations

$$(Q, Q\Theta, Q\Theta^2, Q\Theta^3, Q\Theta^4) \mapsto (Q\Theta, Q\Theta^2, Q\Theta^3, Q\Theta^4, Q),$$

and

$$(Q, Q\Theta, Q\Theta^2, Q\Theta^3, Q\Theta^4) \mapsto (Q, Q\Theta^2, Q\Theta^4, Q\Theta, Q\Theta^3).$$

The second computation follows from the formulae

$$\Theta\chi = \varphi\Theta^{-1}, \quad \Theta^2\chi = \chi\varphi^2\Theta^3, \quad \Theta^3\chi = \varphi^2\Theta, \quad \Theta^4\chi = \varphi\chi\Theta^2.$$

These permutations are all even; hence the permutation group is the alternating group on five letters. Consequently the icosahedral group is simple, and coincides with the group provisionally called the icosahedral group in the earlier abstract discussion.

§60. The Basic Forms of the Icosahedral Group

We have already found one of the three basic forms of the icosahedral group, namely

$$f = x_1x_2(x_1^{10} + 11x_1^5x_2^5 - x_2^{10}). \quad (64)$$

The two missing forms have degrees twenty and thirty.

The basic form of degree twenty is the Hessian covariant of f . We put

$$H = \frac{1}{121} \left(f''(x_1, x_1)f''(x_2, x_2) - f''(x_1, x_2)^2 \right). \quad (65)$$

A straightforward calculation gives

$$H = -(x_1^{20} + x_2^{20}) + 228(x_1^{15}x_2^5 - x_1^5x_2^{15}) - 494x_1^{10}x_2^{10}. \quad (66)$$

The basic form of degree thirty may be defined as the functional determinant of H and f . Set

$$T = \frac{1}{20} (f'(x_1)H'(x_2) - f'(x_2)H'(x_1)). \quad (67)$$

Then

$$T = (x_1^{30} + x_2^{30}) + 522(x_1^{25}x_2^5 - x_1^5x_2^{25}) - 10005(x_1^{20}x_2^{10} + x_1^{10}x_2^{20}). \quad (68)$$

These three forms satisfy an identical relation. To find it, put

$$\lambda = x_1^{10} + x_2^{10}, \quad \mu = x_1^5x_2^5.$$

Then

$$f^5 = \mu(\lambda + 11\mu)^5,$$

and after extracting the common factor $x_1^{10} + x_2^{10}$ from T and forming T^2 , a numerical calculation gives

$$T^2 + H^3 = 1728f^5. \quad (69)$$

This is the required relation.

The icosahedral group is simple, hence by the preceding theory it has no relative invariants. Thus the forms f, H, T remain absolutely unchanged when subjected to the binary icosahedral substitutions written with determinant one. This can also be checked directly, without using the general theorem.

Relation (??) may be written as

$$\frac{T^2}{f^5} + \frac{H^3}{f^5} = 1728. \quad (70)$$

The two quotients T^2/f^5 and H^3/f^5 may be regarded as functions of the single variable $x = x_1 : x_2$. If an icosahedral substitution is applied to this variable, the quotients can change only by constant factors. If

$$\frac{T^2}{f^5}(y) = h \frac{T^2}{f^5}(x), \quad \frac{H^3}{f^5}(y) = k \frac{H^3}{f^5}(x),$$

then, because both x and y are independent variables, the identity (??) gives

$$h \frac{T^2}{f^5} + k \frac{H^3}{f^5} = 1728.$$

Since the two quotients are not in a constant ratio, one must have $h = k = 1$. Hence the quotients remain absolutely unchanged. It follows in turn that f, H, T themselves remain absolutely unchanged under homogeneous icosahedral substitutions of determinant one. For if f were changed into cf , then H and T would be changed into c^2H and c^3T ; since the quotients are unchanged, $c = 1$.

If we set

$$\frac{H^3}{f^5} = z,$$

then (??) gives

$$H^3 - zf^5 = 0, \quad (71)$$

and equivalently

$$T^2 - (1728 - z)f^5 = 0. \quad (72)$$

These are two different forms of the same equation of degree sixty in x , called the *icosahedral equation*. Its properties and its relation to the general equation of the fifth degree will be treated later.

§61. The Invariants of the Icosahedron

The basic forms of the polyhedral groups exhaust, as we now show, the invariant theory of the corresponding groups. We give the proof only for the icosahedral group. The proofs for the other groups are entirely analogous and may be supplied by the reader. The fact that, in the tetrahedral and octahedral groups, relative invariants occur besides absolute invariants, whereas the icosahedral group, being simple, has only absolute invariants, has no essential influence on the argument.

We prove that every invariant of the icosahedral group can be represented as an integral rational function of the three forms f, H, T .

First:

1. No two of the icosahedral forms f, H, T have a common divisor.

This follows at once from the definition of these functions: the equations $f = 0$, $H = 0$, and $T = 0$ give respectively the fivefold, threefold, and twofold poles, and these poles are all distinct.

2. A double root of the icosahedral equation

$$H^3 - zf^5 = 0 \tag{73}$$

is necessarily a root of f , H , or T . Hence it can occur only for one of the values $z = 0, \infty, 1728$.

Indeed, if (??) has a double root, then the function and its first derivative vanish together. In homogeneous notation, by Euler's theorem, the two partial derivatives with respect to x_1 and x_2 vanish simultaneously:

$$\begin{aligned} 3H^2H'(x_1) - 5zf^4f'(x_1) &= 0, \\ 3H^2H'(x_2) - 5zf^4f'(x_2) &= 0. \end{aligned}$$

Since H and f cannot vanish simultaneously, there are only three possibilities: $H = 0$, or $f = 0$, or

$$H'(x_1)f'(x_2) - H'(x_2)f'(x_1) = 0,$$

i.e. $T = 0$, which corresponds by §60 to $z = 1728$.

3. If $J(x_1, x_2)$ is any invariant form of the icosahedral group and ξ is one of its roots, so that $J(\xi, 1) = 0$, then all values obtained from ξ by the fractional icosahedral substitutions are also roots of J .

Consequently, if J has a common divisor with one of the functions f, H, T , then J is divisible by that corresponding form.

Remove from J the highest possible powers of the three basic forms f, H, T , so that the remaining invariant is relatively prime to all three. If ξ is then a root of the remaining form, choose z so that ξ is also a root of (??). By item 2, ξ is a simple root of that equation. By item 3 all of its icosahedral transforms are roots of J , so J is divisible by the left side of (??). The quotient is again an invariant form, and the argument repeats.

We arrive at the theorem:

4. Every invariant of the icosahedron can be represented in the form

$$J(x_1, x_2) = Cf^\alpha H^\beta T^\gamma F(H^3, f^5), \tag{74}$$

where α, β, γ are non-negative integers, C is a constant, and F is a homogeneous integral function.

§62. Polyhedral Groups of the Second Kind. Crystallographic Groups

In §51 we already drew attention to groups of linear ternary substitutions which contain not only proper but also improper orthogonal substitutions. We called them groups of the second kind. We shall now consider the finite ones among them as polyhedral groups of the second kind, or extended polyhedral groups. The substitutions in them with determinant -1 will be called substitutions of the second kind.

According to §50 these groups are isomorphic to groups of binary linear substitutions with determinant $+1$ and -1 . For the fractional substitutions which led us to the polyhedral groups, however, the two kinds cannot be distinguished.

The finite groups of the second kind are important, apart from their general group-theoretic interest, because of their application to crystallography. Their complete determination now presents no essential difficulty. By substitutions we shall here simply mean binary linear substitutions of determinant ± 1 , remembering that two such substitutions whose coefficients differ only by a common sign, such as

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad \begin{pmatrix} -a & -b \\ -c & -d \end{pmatrix},$$

are not to be considered distinct.

Every group of linear substitutions contains the identity, whose determinant is $+1$. Therefore every group Q of the second kind contains, besides improper substitutions, proper substitutions; and these proper substitutions themselves form a group P . If φ is any substitution of the second kind contained in Q , then the product of φ with any other substitution of the second kind in Q lies in P . Hence

$$Q = P + P\varphi. \quad (75)$$

The group P must be one of the polyhedral groups considered above. Moreover

$$\varphi^{-1}P\varphi = P,$$

so that P is a normal divisor of Q . Conversely, if P is a polyhedral group and φ is a substitution of the second kind satisfying $\varphi^{-1}P\varphi = P$, then $P + P\varphi$ is a group of the second kind.

As before, two groups which are transformed into one another are not considered essentially different. We must run through the cases.

I. The identity group

If P is the identity group, viewed as the cyclic group of first degree C_1 , then P contains only the identity. Hence $\varphi^2 = 1$, which is also sufficient. By transformation φ may be brought into the form

$$\begin{pmatrix} a & b \\ c & -a \end{pmatrix},$$

with the condition $a^2 + bc = 1$. This gives two essentially different forms:

$$\varphi' = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \quad \varphi'' = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (76)$$

The first is called inversion, the second reflection. Translated into rectangular coordinates by §50, φ' changes the signs of all three coordinates, whereas φ'' changes x into $-x$. Thus the two groups of the second kind are

$$C'_1 = C_1 + C_1\varphi', \quad C''_1 = C_1 + C_1\varphi''.$$

II. Cyclic groups

Let $P = C_n$, $n > 1$. Put

$$c = \begin{pmatrix} \varepsilon & 0 \\ 0 & \varepsilon^{-1} \end{pmatrix}, \quad \varepsilon = e^{\pi i/n}. \quad (77)$$

Then, as in §55,

$$C_n = 1, c, c^2, \dots, c^{n-1}. \quad (78)$$

If

$$\varphi = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad ad - bc = -1, \quad (79)$$

then $\varphi^{-1}c\varphi$ and $\varphi^{-1}d\varphi$ must occur in the row (??). This forces either $b = c = 0$ or $a = d = 0$. In the first case one obtains two possible extensions,

$$C'_n = \left\{ \begin{pmatrix} e^{\pi i h/n} & 0 \\ 0 & e^{-\pi i h/n} \end{pmatrix} : h = 0, 1, \dots, 2n-1 \right\}, \quad (80)$$

and

$$C''_n = \left\{ \pm \begin{pmatrix} e^{\pi i h/n} & 0 \\ 0 & e^{-\pi i h/n} \end{pmatrix} : h = 0, 1, \dots, n-1 \right\}. \quad (81)$$

For even n both C'_1 and C''_1 occur in C'_n ; for odd n , one of them occurs in C'_n and the other in C''_n . In the second case $a = d = 0$, a transformation which leaves P unchanged reduces φ to

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

producing a third cyclic extension.

III. Dihedral groups

Let $P = D_n$, consisting of

$$D_n = 1, c, c^2, \dots, c^{n-1}, d, cd, c^2d, \dots, c^{n-1}d, \quad d = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}. \quad (82)$$

The substitution c has order n , and the substitutions $c^h d$ all have order two. If $n > 2$, the transform $\varphi^{-1}c\varphi$ must again be among the powers of c , so the preceding analysis applies. Since d does not occur in the cyclic extensions just described, the dihedral groups of the second kind have the forms

$$C'_n + C'_n d, \quad C''_n + C''_n d, \quad C'''_n + C'''_n d. \quad (83)$$

Written out, the first two are the families

$$\begin{pmatrix} e^{\pi i h/n} & 0 \\ 0 & e^{-\pi i h/n} \end{pmatrix}, \quad \begin{pmatrix} 0 & e^{\pi i h/n} \\ e^{-\pi i h/n} & 0 \end{pmatrix}, \quad h = 0, 1, \dots, 2n-1,$$

and the analogous signed family with $h = 0, 1, \dots, n-1$. The third coincides with one of these forms according as n is odd or even.

The case $n = 2$ gives only an apparent exception. One might have $\varphi^{-1}c\varphi = d$ or cd . Direct calculation gives two groups

$$1, c, d, cd, \varphi_1, \varphi_1 c, \varphi_1 d, \varphi_1 cd, \quad 1, c, d, cd, \varphi_2, \varphi_2 c, \varphi_2 d, \varphi_2 cd, \quad (84)$$

where

$$\varphi_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad \varphi_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} i & i \\ i & -i \end{pmatrix}.$$

The formulae

$$c\varphi_1 = \varphi_1 d, \quad d\varphi_1 = \varphi_1 c, \quad cd\varphi_1 = \varphi_1 dc,$$

and their analogues for φ_2 , show that these groups are transformed forms of D_2 .

For the remaining polyhedral groups of the second kind, two general remarks are useful. The inversion

$$j = \begin{pmatrix} i & 0 \\ 0 & i \end{pmatrix} \quad (85)$$

is a similarity substitution and therefore commutes with every substitution. Thus from every polyhedral group of the first kind one obtains at least one group of the second kind,

$$P + Pj. \quad (86)$$

By Sylow's theorem, every finite group contains a subgroup whose order is the highest power of two dividing the order of the group. The extended tetrahedral, octahedral, and icosahedral groups have orders 24, 48, 120; hence they must contain divisors of order 8, 16, 8. In the tetrahedral and icosahedral groups there is a four-group D_2 but no substitution of fourth order; in the octahedral group there is a dihedral group D_4 but no substitution of eighth order. Hence one of the extended dihedral groups just considered must occur.

For the tetrahedral group two extensions occur. Besides

$$T' = T + Tj,$$

one obtains another

$$T'' = T + Tj_1,$$

where j_1 is the second extension of the four-group. No further tetrahedral extension occurs, because the tetrahedral group is completely determined by any one four-group contained in it. If an extension of the four-group of one of the exceptional forms (??) is chosen, the resulting tetrahedral group is merely a transform of T'' .

For the octahedral group, the special substitution defining the second tetrahedral extension does not commute with the octahedral group, and consequently the octahedral group gives no further extension beyond inversion:

$$O' = O + Oj.$$

The same is true of the icosahedral group. One may transform the icosahedral group so that one of its four-groups has the simplest form; the calculation is longer but entirely straightforward, and leads to no additional case. Thus one obtains

$$J' = J + Jj.$$

The following groups of the second kind are therefore obtained:

$$\begin{array}{l} C'_1, C''_1, \quad C'_n, C''_n, C'''_n, \quad D'_n, D''_n, D'''_n, \\ T', T'', \quad O', \quad J'. \end{array}$$

The thirty-two symmetry systems of crystallographers are contained among them. In the crystallographic notation they are the groups

$$\begin{array}{l} C_1, \quad C'_1, \quad C''_1, \quad D_1, \\ C_2, \quad C'_2, \quad C''_2, \quad D_2, \quad D'_2, \quad D''_2 \\ C_3, \quad C'_3, \quad C''_3, \quad D_3, \quad D'_3, \quad D''_3 \\ C_4, \quad C'_4, \quad C''_4, \quad D_4, \quad D'_4, \\ C_6, \quad C'_6, \quad C''_6, \quad D_6, \quad D'_6, \\ T, \quad T', \quad T'', \quad O, \quad O'. \end{array}$$

The remaining polyhedral groups are excluded in crystallography by the crystallographic law of rational indices.

§63. Function Congruences

From the linear substitutions whose construction and composition we have studied in the preceding sections one can derive another kind of finite group, namely the congruence groups.

These groups are of many-sided interest. They arise first in the theory of elliptic functions, where they appear as Galois groups of modular equations. They are also important for general group theory, because they furnish a method for constructing whole series of simple groups. This is all the more noteworthy since, as we saw in the fourth section, simple groups are rather rare, at least among the lower degrees.

The theory of these congruence groups is not only greatly generalized, but the governing laws appear much more sharply, if one bases it on an extension of the notion of congruence due to Gauss and subsequently developed and applied, especially by Galois, in problems of group theory.

To obtain a basis for the theory, consider an integral polynomial in one variable t ,

$$f(t) = a_0 t^n + a_1 t^{n-1} + \cdots + a_{n-1} t + a_n, \quad (87)$$

with integral coefficients. Choose a prime number p as modulus, and assume that a_0 is not divisible by p .

Two integral functions of t whose corresponding coefficients are congruent modulo p will themselves be called congruent modulo p . The function $f(t)$ is called reducible modulo p if there exist two integral functions $\varphi(t), \psi(t)$, each with at least one term depending on t whose coefficient is not divisible by p , such that

$$f(t) \equiv \varphi(t)\psi(t) \pmod{p}. \quad (88)$$

Terms with coefficients divisible by p may of course be omitted from φ and ψ , and then the degree of their product equals the degree of f . Hence the degrees of φ and ψ are both lower than n . If no such functions exist, $f(t)$ is called irreducible modulo p .

A function irreducible modulo p is certainly irreducible over the field of rational numbers. The converse, however, need not hold. For example, a quadratic function $t^2 - B$ is reducible modulo p when B is a quadratic residue of p , for if $a^2 \equiv B \pmod{p}$, then

$$t^2 - B \equiv (t - a)(t + a) \pmod{p}.$$

On the other hand, $t^2 - N$ is irreducible when N is a non-residue of p , since otherwise $t^2 - N$ would vanish modulo p for some rational integer t .

The functions

$$t^2 + t + 1, \quad t^3 + t + 1$$

for the modulus two give another example. They are irreducible modulo two. If either were reducible, one factor would have to be linear, and there would have to be an integer which, substituted for t , makes the function even. This is impossible, since for $t = 0$ and $t = 1$ both functions take odd values.

Let

$$P = t^n + p_1 t^{n-1} + p_2 t^{n-2} + \cdots + p_{n-1} t + p_n \quad (89)$$

be irreducible modulo p , and for simplicity take the leading coefficient to be one. From any other integral function $F(t)$ one may derive a remainder $\Phi(t)$ of degree lower than n by writing

$$F(t) = QP + \Phi(t), \quad (90)$$

where Q has integral coefficients. We describe the relation of F to Φ as a congruence modulo P .

Thus two functions $F(t), F_1(t)$ which have the same remainder modulo P are called congruent modulo P , and this is expressed by

$$F(t) \equiv F_1(t) \pmod{P}.$$

Equivalently, $F(t) - F_1(t)$ is divisible by P .

If the two functions do not have the same remainder, but have remainders congruent modulo p , then the functions are called congruent modulo the double modulus P, p :

$$F(t) \equiv F_1(t) \pmod{P, p}. \quad (91)$$

For this kind of congruence one has:

1. The product of two functions cannot be congruent to zero unless one of the factors is congruent to zero.

The proof follows from the algorithm for the greatest common divisor. Suppose F_1 has degree lower than P and is not congruent to zero modulo p . One can determine functions $F_2, F_3, \dots$, also of lower degree than P , of decreasing degree, and quotients $Q_1, Q_2, \dots$, such that

$$P = Q_1 F_1 + F_2, \quad F_1 = Q_2 F_2 + F_3, \quad \dots, \quad F_{r-2} = Q_{r-1} F_{r-1} + F_r \pmod{p}, \quad (92)$$

until the last remainder F_r is a constant. This constant cannot be congruent to zero modulo p ; otherwise P would be reducible modulo p . If now $Q F_1 \equiv 0 \pmod{P, p}$, multiplication of the equations (??) by Q gives successively $Q F_2 \equiv 0, \dots, Q F_r \equiv 0$, hence $Q \equiv 0$. Taking F_1, Q to be the remainders of the two factors proves the assertion.

The remainders of all functions modulo P have the form

$$X = X(t) = x_0 + x_1 t + \dots + x_{n-1} t^{n-1}. \quad (93)$$

To obtain one representative from each congruence class modulo P, p , it is enough to let each coefficient $x_0, x_1, \dots, x_{n-1}$ run through the residues $0, 1, \dots, p-1$. Thus there are exactly p^n incongruent functions modulo P, p .

§64. Congruence Fields

It is only a different mode of expression for the preceding considerations to introduce, with Galois, an imaginary number ε satisfying

$$P(\varepsilon) \equiv 0 \pmod{p}. \quad (94)$$

No ordinary rational integer of this sort exists when $n > 1$, for then $P(t)$ would be divisible by $t - \varepsilon$ and so would be reducible. Nevertheless one may use such a symbol in calculation, just as one introduces the imaginary unit $i = \sqrt{-1}$ in ordinary arithmetic. As far as addition, subtraction, and multiplication are concerned, one calculates with it according to the ordinary rules of algebra, using the equation $P(\varepsilon) = 0$ whenever desired. One may even think of ε as an actual root of $P(x) = 0$ in the usual sense.

All numbers obtained in this way can be written in the form

$$\alpha = a_0 + a_1 \varepsilon + \dots + a_{n-1} \varepsilon^{n-1}. \quad (95)$$

Because the modulus p is held fixed, two numbers congruent modulo p will simply be regarded as equal. Strictly speaking, therefore, the objects of the calculation are not the individual numbers but the classes of numbers congruent to one another. These numbers α are called the Galois imaginaries. The system associated with a fixed ε will be denoted by $\mathfrak{G}$.

We have first:

2. There are exactly p^n different numbers in $\mathfrak{G}$.

From item 1 of the preceding section we also obtain:

3. A product of two or more numbers of $\mathfrak{G}$ is zero if and only if at least one factor is zero.

The whole system $\mathfrak{G}$ is obtained by allowing the rational coefficients $a_0, a_1, \dots, a_{n-1}$ in (??) to run independently through a complete residue system modulo p . It contains the ordinary residue classes $0, 1, \dots, p-1$, and for $n = 1$ it is identical with that system.

If $\alpha \neq 0$ is fixed in $\mathfrak{G}$, then $\alpha\xi$ runs through the whole system at the same time as ξ . Indeed, $\alpha\xi = \alpha\xi'$ implies $\xi = \xi'$. Hence:

4. If $\alpha, \beta \in \mathfrak{G}$ and $\alpha \neq 0$, there exists exactly one $\gamma \in \mathfrak{G}$ such that

$$\alpha\gamma = \beta.$$

This proves that division, except by zero, is allowed in $\mathfrak{G}$. We write this γ as β/α .

The system $\mathfrak{G}$ may be called a finite field: it contains only finitely many numbers, and it has the characteristic property of a field, namely the unrestricted performance of the arithmetic operations, except division by zero. We shall call $\mathfrak{G}$ a *congruence field*. The integer n is its degree and p its modulus.

If $\alpha \neq 0$, then $\alpha\xi = \eta$ runs through the non-zero elements as ξ runs through them. Multiplying all such equations gives

$$\alpha^{p^n-1}\Pi = \Pi,$$

where Π denotes the non-zero product of all non-zero elements. Cancelling Π yields Fermat's theorem in this field:

$$\alpha^{p^n-1} = 1. \tag{96}$$

Multiplication by α gives the equivalent form

$$\alpha^{p^n} = \alpha, \tag{97}$$

which also holds for $\alpha = 0$.

If $\alpha \neq 0$, let e be the least positive exponent with

$$\alpha^e = 1. \tag{98}$$

Then the powers $1, \alpha, \alpha^2, \dots, \alpha^{e-1}$ are distinct, and α is said to belong to the exponent e . The exponent e divides $p^n - 1$. For if $\alpha^m = 1$ and $m = qe + r$ with $0 \leq r < e$, then $\alpha^r = 1$, hence $r = 0$. In particular $e \mid p^n - 1$.

If $p^n - 1 = ef$, then α^h belongs to the exponent e precisely when h is relatively prime to e . Thus, if there is any element with exponent e , there are $\varphi(e)$ of them, where φ is Euler's function. Since

$$\sum_{e \mid (p^n-1)} \varphi(e) = p^n - 1, \tag{99}$$

and since every non-zero element belongs to exactly one exponent, it follows that, for every divisor e of $p^n - 1$, there are exactly $\varphi(e)$ elements of exponent e . In particular, there are $\varphi(p^n - 1)$ elements γ whose powers

$$1, \gamma, \gamma^2, \dots, \gamma^{p^n-2} \tag{100}$$

are all distinct. Such elements are called primitive roots of $\mathfrak{G}$.

Among the elements of $\mathfrak{G}$ some are squares, and the others are non-squares. If $p = 2$, then every element is a square, as follows from (??). If p is odd, then one half of the non-zero elements

are squares and the other half are non-squares. Indeed, the two opposite elements α and $-\alpha$ are distinct but have the same square, and $\beta^2 = \alpha^2$ implies $\beta = \pm\alpha$. Thus there are exactly $(p^n - 1)/2$ non-zero squares and the same number of non-squares. Counting zero among the squares, one has:

5. If $p = 2$, all elements of $\mathfrak{G}$ are squares. If p is odd, there are $(p^n + 1)/2$ squares and $(p^n - 1)/2$ non-squares.

For odd p , a primitive root is necessarily a non-square, and in the sequence (??) the even powers are the squares and the odd powers the non-squares.

The product and quotient of two squares is again a square. Hence the product of a non-square and a non-zero square is a non-square, and the product of two non-squares is a square. All these assertions also follow immediately from the representation by powers of a primitive root.

We close with a result needed later:

6. Every non-square is the sum of two squares.

Here p is assumed odd, since otherwise there are no non-squares. First, every element β can be represented as a difference of two squares, because

$$\beta = \left(\frac{\beta+1}{2}\right)^2 - \left(\frac{\beta-1}{2}\right)^2.$$

If -1 is a square in $\mathfrak{G}$, this is already a representation of β as a sum of two squares. If -1 is a non-square, choose in the natural sequence $1, 2, \dots, p-1$ a square immediately followed by a non-square. Thus let $a = \xi^2$ be a square and $\xi^2 + 1$ a non-square. If β is a non-square, then $\beta/(\xi^2 + 1) = \eta^2$ is a square, and so

$$\beta = (\xi\eta)^2 + \eta^2.$$

§65. Congruence Groups in the Field $\mathfrak{G}$

In every congruence field $\mathfrak{G}$ one can form a finite group of linear substitutions

$$A = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \tag{101}$$

by letting $\alpha, \beta, \gamma, \delta$ run through all elements of $\mathfrak{G}$, and by composing two substitutions according to

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \begin{pmatrix} \alpha' & \beta' \\ \gamma' & \delta' \end{pmatrix} = \begin{pmatrix} \alpha\alpha' + \beta\gamma' & \alpha\beta' + \beta\delta' \\ \gamma\alpha' + \delta\gamma' & \gamma\beta' + \delta\delta' \end{pmatrix}. \tag{102}$$

The degree of this full group is p^{4n} when p is the modulus and n the degree of $\mathfrak{G}$.

It contains a group of smaller degree. Let

$$\det A = \alpha\delta - \beta\gamma.$$

By (??), determinants multiply. Therefore, if the entries are restricted by

$$\alpha\delta - \beta\gamma = 1, \tag{103}$$

the corresponding substitutions form a group. Its degree is the number of solutions of (??). To count them, first choose α, β not both zero; this gives $p^{2n} - 1$ choices. If a particular solution (γ_0, δ_0) is known, all others have the form

$$\gamma = \gamma_0 + \alpha\xi, \quad \delta = \delta_0 + \beta\xi,$$

with $\xi \in \mathfrak{G}$. Thus there are p^n choices for (γ, δ) , and the degree of the determinant-one group is

$$p^n(p^{2n} - 1). \quad (104)$$

For odd p one can derive a still smaller group. If $p = 2$, a number α is not distinct from $-\alpha$. If p is odd, however, the two matrices

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}, \quad \begin{pmatrix} -\alpha & -\beta \\ -\gamma & -\delta \end{pmatrix} \quad (105)$$

are distinct, and the whole system splits into such pairs. Multiplication of pairs again gives a pair. Hence these pairs can themselves be regarded as the elements of a new group of degree

$$\frac{1}{2}p^n(p^{2n} - 1). \quad (106)$$

This group, of degree (??) for $p = 2$ and (??) for odd p , will be called the congruence group belonging to $\mathfrak{G}$, and denoted by E . In modern notation it is the projective special linear group over the finite field $\mathfrak{G}$.

For further study it is important to know generators. Write the elements of $\mathfrak{G}$ in the form

$$\xi = x_0 + x_1\varepsilon + \cdots + x_{n-1}\varepsilon^{n-1}, \quad (107)$$

where the x_i are rational residues modulo p . A generating system is

$$A_k = \begin{pmatrix} 1 & \varepsilon^k \\ 0 & 1 \end{pmatrix}, \quad B_k = \begin{pmatrix} 1 & 0 \\ \varepsilon^k & 1 \end{pmatrix}, \quad k = 0, 1, \dots, n-1. \quad (108)$$

Indeed,

$$\begin{pmatrix} 1 & \alpha \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & \beta \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & \alpha + \beta \\ 0 & 1 \end{pmatrix},$$

and repeated use gives every upper unipotent matrix; similarly every lower unipotent matrix is generated. Combining such elements gives matrices of the forms

$$\begin{pmatrix} \alpha & 1 \\ -1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & \beta \\ 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ \gamma & 1 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

and hence, by a further multiplication,

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \quad (109)$$

whenever $\delta \neq 0$. The restriction $\delta \neq 0$ may be removed using the last displayed matrix. Thus the elements (??) generate E .

The group E contains a divisor of index $p^n + 1$, consisting of all substitutions

$$\begin{pmatrix} \alpha & \beta \\ 0 & \delta \end{pmatrix}, \quad \alpha\delta = 1, \quad (110)$$

where β is arbitrary and $\alpha \neq 0$. This divisor yields a permutation representation of E on $p^n + 1$ letters.

The linear expression

$$\eta = \frac{\alpha\xi + \beta}{\gamma\xi + \delta} \quad (111)$$

assigns to every $\xi \in \mathfrak{G}$ a corresponding η , except when $\gamma\xi + \delta = 0$. Conversely, each η determines a ξ , except when $\gamma\eta - \alpha = 0$. To remove the exceptional cases, add to $\mathfrak{G}$ one new element, called infinity and denoted by ∞ . It is governed by the usual rules

$$a + \infty = \infty, \quad a \cdot \infty = \infty \quad (a \neq 0), \quad \frac{a}{0} = \infty, \quad \frac{a}{\infty} = 0.$$

Then every substitution of E acts as a permutation of the $p^n + 1$ symbols of the projective line $\mathfrak{G} \cup \{\infty\}$.

§66. Simplicity of the Group E

The first question in a closer investigation of the group E is whether it has a normal divisor. One can prove that, apart from two quite small exceptional cases, none exists.

Assume that G is a normal divisor of E and that it contains at least one substitution different from the identity,

$$A = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}. \quad (112)$$

By normality, for every element $T \in E$ the transform

$$TAT^{-1} \quad (113)$$

also lies in G . We shall show that by such transformations and their products one can derive all elements of E . It is enough to show that the generators A_k, B_k of (??) lie in G .

We proceed step by step.

First, G contains a substitution of the form

$$\begin{pmatrix} 1 & \lambda \\ 0 & 1 \end{pmatrix}. \quad (114)$$

To see this, form the transform of A by $\begin{pmatrix} 1 & \xi \\ 0 & 1 \end{pmatrix}$. If $\beta \neq 0$, choose ξ so that the lower-left entry vanishes. If $\beta = 0$, then not all of γ and $\delta - \alpha$ vanish, since otherwise A would be the identity; choose ξ so that the transformed matrix has non-zero upper-right entry, and apply the preceding argument once more.

Second, G contains a substitution of the form

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \quad (115)$$

Weber obtains this by solving, for λ, μ, ν, ρ , the system

$$\lambda\rho - \mu\nu = 1, \quad \mu\rho\gamma - \lambda\nu\beta - \lambda\nu\delta = 0, \quad -\mu^2\gamma + \lambda\mu\beta + \lambda\mu\delta = 1. \quad (116)$$

From the last two equations one derives

$$\lambda\beta + \mu\delta = 0, \quad \mu\gamma = \nu, \quad (117)$$

and hence the remaining condition

$$\lambda^2\beta + \lambda\mu\delta - \mu^2\gamma = 1. \quad (118)$$

This equation can always be solved. If β is a square, put $\beta = w^2$ and determine λ . This includes the case $p = 2$. If p is odd and β is a non-square, one reduces the problem to the decomposition of a non-square as a sum of two squares, supplied by §64. Thus the desired transformation exists.

Third, except in the case $n = 1, p = 2$, the group G contains

$$A_0 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}. \quad (119)$$

Indeed, from the preceding step the group contains

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

and hence also the products

$$\begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ -1 & 2 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ -1 & m \end{pmatrix}$$

for every odd integer m . If p is odd, take $m = p$ and obtain A_0 . If $p = 2$, a further calculation gives, for every non-zero $q \in \mathfrak{G}$, a matrix of the form

$$\begin{pmatrix} 0 & q \\ q^{-1} & 0 \end{pmatrix},$$

and products of such matrices yield A_0 provided $n > 1$. The only exception is $n = 1, p = 2$, where one obtains a group of degree six with a normal divisor of degree three.

Fourth, except in the case $n = 1, p = 3$, the group G contains every substitution

$$\begin{pmatrix} 1 & \xi \\ 0 & 1 \end{pmatrix} \quad (120)$$

with arbitrary $\xi \in \mathfrak{G}$. For, using A_0 , one obtains

$$\begin{pmatrix} 1 & (q^2 - 1)\lambda \\ 0 & 1 \end{pmatrix} \quad (121)$$

for every non-zero q . If one can choose q so that $q^2 - 1 \neq 0$, then λ can be adjusted to give any ξ . With a primitive root γ of $\mathfrak{G}$, put $q = \gamma^h$. Such a choice is always possible except in the small cases $n = 1, p = 3$ and $n = 1, p = 5$. The latter is removed by an explicit calculation using the matrix

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} -2 & -1 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} -2 & 0 \\ 0 & 2 \end{pmatrix},$$

which again yields all elements (??).

Thus in all cases except $n = 1, p = 2$ and $n = 1, p = 3$, all generators of E lie in G , hence $G = E$. The group E is therefore simple.

For the exceptional case $n = 1, p = 3$, the group E has degree twelve and contains the normal divisor of degree four

$$G = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \begin{pmatrix} -1 & 1 \\ -1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix} \right\}. \quad (122)$$

The remaining cosets verify directly that this is normal. For $n = 1, p = 2$, the group is isomorphic to the symmetric group on three letters and has its usual normal subgroup of order three.

For the first non-exceptional cases one obtains the degrees

$$n = 1, p = 5 : \quad g = 60,$$

and then the higher degrees supplied by $\frac{1}{2}p^n(p^{2n} - 1)$ for odd p , and by $p^n(p^{2n} - 1)$ for $p = 2$.

§67. Congruence Fields of Degree Two

Every congruence field $\mathfrak{G}_{n,p}$ of degree n and modulus p contains, as a subfield, the field $\mathfrak{G}_{1,p}$ consisting of rational residues modulo p . Consequently the group of linear substitutions $E_{n,p}$ contains as a divisor the group of all substitutions

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad (123)$$

where a, b, c, d are rational residues modulo p and $ad - bc = 1$, again with opposite matrices identified when p is odd.

This group is the group which appears in the theory of elliptic functions as the group of modular equations. It is the real goal of the present investigation. For the study of this group, especially for the search for its divisors, it is most advantageous not to treat it by itself, but to regard it as a divisor of a group $E_{2,p}$.

Let N be a fixed quadratic non-residue modulo p . The polynomial $t^2 - N$ is then irreducible modulo p . We obtain a congruence field of second degree by adjoining a Galois imaginary ε satisfying

$$\varepsilon^2 = N. \quad (124)$$

In what follows, lower-case Latin letters denote rational residues modulo p , which we shall also call real numbers; Greek letters denote elements of the field $\mathfrak{G}_{2,p}$.

The field $\mathfrak{G}_{2,p}$ has the important property that every real number is a square in it. If a is already a quadratic residue modulo p , write $a = x^2$. If b is a non-residue, then $b = aN = (x\varepsilon)^2$ after multiplying by a suitable square a .

A number of the form $a\varepsilon$ will be called purely imaginary. Two numbers $a + b\varepsilon$ and $a - b\varepsilon$ will be called conjugate imaginary. This transfer of terminology from the ordinary complex numbers is justified by the agreement of the calculation rules and cannot cause confusion, since no ordinary complex numbers are involved here.

Since $N^{(p-1)/2} \equiv -1 \pmod{p}$, equation (??) gives

$$\varepsilon^p = -\varepsilon. \quad (125)$$

For $\alpha = a + b\varepsilon$, the binomial theorem gives, modulo p ,

$$\alpha^p = a^p + b^p \varepsilon^p = a - b\varepsilon. \quad (126)$$

Therefore

$$\alpha^{p+1} = a^2 - Nb^2, \quad (127)$$

which is always real.

If γ is a primitive root of $\mathfrak{G}_{2,p}$, then γ^r is real if and only if r is divisible by $p + 1$. In particular γ^{p+1} is a primitive root of the prime field in the ordinary sense.

The necessary and sufficient condition that $\alpha \in \mathfrak{G}_{2,p}$ be real is

$$\alpha^p = \alpha. \quad (128)$$

Moreover every real number is a $(p + 1)$ -st power in $\mathfrak{G}_{2,p}$. Conversely, a power γ^r is real only when r is divisible by $p + 1$, because every non-zero real number satisfies Fermat's congruence $x^{p-1} = 1$.

The notion of roots of unity also carries over. If m divides $p^2 - 1$ and

$$\rho = \gamma^{(p^2-1)/m},$$

then $\rho^m = 1$, and no lower positive power is one. Thus ρ is a primitive m -th root of unity in $\mathfrak{G}_{2,p}$. All m -th roots of unity are powers ρ^s , and the primitive ones are precisely those with s prime to m . Every non-zero element of $\mathfrak{G}_{2,p}$ is a root of unity of degree $p^2 - 1$.

§68. The Real Linear Congruence Group L_p

Equipped with these tools, we now investigate the real congruence group L_p , consisting of all substitutions

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad ad - bc = 1, \quad (129)$$

where a, b, c, d are real residues modulo p . In the earlier notation this group is $E_{1,p}$. Excluding $p = 2$, its degree is

$$\frac{1}{2}p(p^2 - 1). \quad (130)$$

If A, U are elements of a group, we call

$$U^{-1}AU = A'$$

the element obtained from A by transformation with U . Elements obtained from one another in this way have the same order. If all elements A are taken from a group G , the transformed elements form an isomorphic group $U^{-1}GU$.

Let $A \in L_p$, and let

$$U = \begin{pmatrix} \lambda & \mu \\ \nu & \rho \end{pmatrix}$$

be an element of $E_{2,p}$. We first try to transform A into the normal form

$$S = \begin{pmatrix} \theta & 0 \\ 0 & \theta^{-1} \end{pmatrix}. \quad (131)$$

From the equation

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} \lambda & \mu \\ \nu & \rho \end{pmatrix} = \begin{pmatrix} \lambda & \mu \\ \nu & \rho \end{pmatrix} \begin{pmatrix} \theta & 0 \\ 0 & \theta^{-1} \end{pmatrix}$$

one obtains

$$\begin{aligned} (a - \theta)\lambda + b\nu &= 0, & (a - \theta^{-1})\mu + b\rho &= 0, \\ c\lambda + (d - \theta)\nu &= 0, & c\mu + (d - \theta^{-1})\rho &= 0. \end{aligned} \quad (132)$$

Thus θ and θ^{-1} must be the roots of

$$\theta^2 - \theta(a + d) + 1 = 0. \quad (133)$$

Once θ and θ^{-1} are known, (??) determines the ratios $\nu : \lambda$ and $\rho : \mu$. The scalar factors are still to be chosen so that $\lambda\rho - \mu\nu = 1$. This is possible unless $a + d = \pm 2$, for otherwise the two eigenvectors would be proportional.

The equation (??) is always solvable in $\mathfrak{G}_{2,p}$, since every real number is a square there. Explicitly,

$$\theta = \frac{a + d}{2} \pm \sqrt{\left(\frac{a + d}{2}\right)^2 - 1}. \quad (134)$$

We therefore have:

1. A substitution A is transformable into the normal form S whenever $a + d \neq \pm 2$. The number θ is real or imaginary according as $(a + d)^2 - 4$ is a quadratic residue or a non-residue modulo p .

In the remaining case $a + d = \pm 2$, the normal form S cannot be produced. Instead A can be transformed into

$$T = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \quad (135)$$

with real t . It is enough to take $a + d = 2$, since the case -2 is obtained by changing all signs. Then

$$a - 1 = -(d - 1), \quad (a - 1)(d - 1) = bc. \quad (136)$$

If $b = 0$ or $c = 0$, the transformation is immediate; if both are non-zero, the displayed identities give an explicit transforming matrix. Hence:

2. A substitution A with $a + d = \pm 2$ can always be transformed into the normal form T .

The orders of elements in L_p are now easily determined. There are three cases.

First kind. If $a + d = \pm 2$, every non-identity substitution is transformed into T . Since

$$T^r = \begin{pmatrix} 1 & rt \\ 0 & 1 \end{pmatrix},$$

such a substitution has order p . Counting these elements gives p^2 substitutions of the first kind, including the identity.

Second kind. If $(a + d)^2 - 4$ is a quadratic residue modulo p , the substitution is transformed into S with real θ . Let γ be a primitive root of $\mathfrak{G}_{2,p}$. Then the real elements may be written as powers $\gamma^{r(p+1)}$, and

$$a + d = \gamma^{r(p+1)} + \gamma^{-r(p+1)}. \quad (137)$$

Distinct values, different from ± 2 , are obtained for

$$r = 1, 2, \dots, \frac{p-3}{2}. \quad (138)$$

The order of S , and hence of the substitution, is a divisor of $(p-1)/2$. It is exactly $(p-1)/2$ when r is prime to $(p-1)/2$. The number of substitutions of the second kind is

$$\frac{1}{2}p(p-3)(p+1). \quad (139)$$

Third kind. If $(a + d)^2 - 4$ is a quadratic non-residue, then θ is not real but is conjugate to its reciprocal. Therefore $\theta^{p+1} = 1$, and the order is a divisor of $(p+1)/2$. Write

$$a + d = \gamma^{r(p-1)} + \gamma^{-r(p-1)}, \quad r = 1, 2, \dots, \frac{p-1}{2}. \quad (140)$$

Here neither $b = 0$ nor $c = 0$ can occur; otherwise a and d would be real eigenvalues. Counting gives

$$\frac{1}{2}p(p-1)(p-1) \quad (141)$$

substitutions of the third kind. Together with the numbers from the first and second kinds, this gives the total degree $\frac{1}{2}p(p^2 - 1)$, as required.

§69. Imaginary Form of the Group L_p

The substitutions of the first and second kinds in L_p have the property that their corresponding normal forms T or S lie again in L_p , and that they can be transformed into these forms by real substitutions, i.e. by substitutions lying in L_p itself. This does not hold for the substitutions of the third kind.

We shall transform the whole group L_p into another group A_p , which is an isomorphic divisor of $E_{2,p}$, in such a way that the normal forms of the transformations of the third kind are contained in A_p , and each substitution of the third kind in A_p can be transformed into normal form by substitutions of A_p itself.

Consider

$$S = \begin{pmatrix} \theta & 0 \\ 0 & \theta^{-1} \end{pmatrix}, \quad (142)$$

where θ and θ^{-1} are conjugate imaginary. We seek a substitution

$$R = \begin{pmatrix} \lambda & \mu \\ \nu & \rho \end{pmatrix} \quad (143)$$

so that

$$RSR^{-1} \quad (144)$$

is real. Direct calculation shows that this is achieved when $\lambda\mu$ and $\rho\nu$ are purely imaginary and $\lambda\rho, -\mu\nu$ are conjugate imaginary pairs. A particularly simple choice, with determinant one, is

$$R = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ \varepsilon^{-1} & -\varepsilon^{-1} \end{pmatrix}. \quad (145)$$

If

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

is any real substitution in L_p , then, using $\varepsilon^2 = N$, one obtains

$$R^{-1}AR = \begin{pmatrix} \alpha & \beta \\ N\beta' & \alpha' \end{pmatrix}, \quad (146)$$

where α, α' and β, β' are conjugate imaginary pairs; equivalently $\alpha' = \alpha^p, \beta' = \beta^p$. This may also be written in the form

$$\begin{pmatrix} \alpha & \beta \\ N\beta^p & \alpha^p \end{pmatrix}. \quad (147)$$

Conversely, from every matrix of the form (??) one obtains by the inverse transformation a real substitution in L_p . Thus, as A runs through L_p , the transformed matrices form an isomorphic group, which we denote by A_p .

In A_p the normal form S for substitutions of the third kind is present. It remains to check that any substitution A of the third kind in A_p can be brought to this normal form by transformation with a substitution $U \in A_p$. Write

$$U = \begin{pmatrix} \lambda & \mu \\ N\mu' & \lambda' \end{pmatrix}, \quad \lambda\lambda' + N\mu\mu' = 1. \quad (148)$$

The equations corresponding to §68 become

$$(\theta - \alpha)(\theta - \alpha') + N\beta\beta' = 0, \quad (149)$$

and

$$\frac{N\lambda\mu'}{\theta - \alpha} = \frac{\lambda'\mu}{\theta - \alpha'} = \frac{\beta}{\beta'}. \quad (150)$$

The second equation follows from the first by passing to conjugate imaginaries. Hence one can choose

$$N\lambda\mu' = k(\alpha - \theta), \quad N\lambda'\mu = k'(\alpha' - \theta^{-1}),$$

where k, k' are conjugate imaginary, and then determine k, k' so that $\lambda\lambda' + N\mu\mu' = 1$. This proves the claim.

Let γ be a primitive root of $\mathfrak{G}_{2,p}$. For the normal forms of the second and third kinds we may write

$$S_1 = \begin{pmatrix} \gamma^{p+1} & 0 \\ 0 & \gamma^{-(p+1)} \end{pmatrix}, \quad S_2 = \begin{pmatrix} \gamma^{p-1} & 0 \\ 0 & \gamma^{-(p-1)} \end{pmatrix}. \quad (151)$$

If A is a substitution of the second kind in L_p , and $\bar{A}$ a substitution of the third kind in A_p , then there are substitutions $U \in L_p$ and $V \in A_p$ such that

$$A = US_1^r U^{-1}, \quad \bar{A} = VS_2^r V^{-1}.$$

Writing

$$T_1 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

the same applies to substitutions of the first kind. Hence:

The substitutions of the first, second, and third kinds in L_p or A_p , together with the identity, can be arranged in cycles of respectively

$$p, \quad \frac{p-1}{2}, \quad \frac{p+1}{2}$$

elements.

Two such cycles have no element in common except the identity. For first-kind substitutions this is immediate, since a whole cycle is generated by any non-identity element in it. For the second and third kinds, if two cycles have an element in common, transform one of them so that the common element is in normal form:

$$1, USU^{-1}, US^2U^{-1}, \dots; \quad 1, S, S^2, \dots \quad (152)$$

If for two non-zero exponents

$$US^rU^{-1} = S^s,$$

then calculation forces either U to be diagonal, giving $USU^{-1} = S$, or anti-diagonal, giving $USU^{-1} = S^{-1}$. In both cases the two cycles coincide.

From the enumeration in §68, the numbers of cycles of the first, second, and third kinds are therefore

$$p+1, \quad \frac{1}{2}p(p+1), \quad \frac{1}{2}p(p-1). \quad (153)$$

It is useful, besides L_p and A_p , to consider a third group in $E_{2,p}$, isomorphic to them but not transformable into them within the same field. Since N is real, choose $v \in \mathfrak{G}_{2,p}$ with

$$N = v^{p+1}.$$

To the substitution

$$A = \begin{pmatrix} \alpha & \beta \\ N\beta^p & \alpha^p \end{pmatrix}$$

of A_p let correspond

$$B = \begin{pmatrix} \alpha & v\beta \\ -v^p\beta^p & \alpha^p \end{pmatrix}, \quad \alpha\alpha^p + N\beta\beta^p = 1. \quad (154)$$

The product of two such matrices again has the same form and corresponds to the product of the corresponding A 's. Thus the totality of matrices B forms a group isomorphic to A_p , denoted by F_p .

Replacing $v\beta$ by β gives the simpler form

$$B = \begin{pmatrix} \alpha & \beta \\ -\beta^p & \alpha^p \end{pmatrix}, \quad \alpha\alpha^p + \beta\beta^p = 1. \quad (155)$$

To pass from such a B to the corresponding real substitution A , first form

$$\bar{A} = \begin{pmatrix} \alpha & v^{-1}\beta \\ N(v^{-1}\beta)^p & \alpha^p \end{pmatrix}, \quad (156)$$

and then transform back by

$$A = R\bar{A}R^{-1}. \quad (157)$$

Although F_p is isomorphic to L_p and A_p , it cannot be transformed into either by substitutions whose entries belong to $\mathfrak{G}_{2,p}$. To perform such a transformation one would have to pass to a higher field in which all elements of $\mathfrak{G}_{2,p}$ become squares.